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\\Research\Investigators\Frederick\allcustomer_20260909\can_neuromelanin
TA: 14 sec Coil Selection: Auto Voxel Size: 2.3×2.3×5.0 mm³ Acc:: None Rel. SNR: 1.00

Properties

Start measurement without further preparation	On
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
Graphic segment	Default
Inline Movie	Off

Routine

Slice Group	1
Slices	1
Distance Factor	20 %
Position	Isocenter
Orientation	Transversal
Phase Encoding Dir.	A >> P
Phase Oversampling	0 %
FOV Read	300 mm
FOV Phase	100.0 %
Slice Thickness	5.0 mm
TR	100.0 ms
TE	10.00 ms
Averages	1
Concatenations	1
AutoAlign	---

Contrast - Common

TR	100.0 ms
TE	10.00 ms
MTC	Off
Magn. Preparation	None
Flip Angle	25 deg
Fat-Water Contrast	Standard
Dark Blood	Off
Contrasts	1
SWI	Off
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	1
Multiple Series	Each Measurement

Resolution - Common

FOV Read	300 mm
FOV Phase	100.0 %
Slice Thickness	5.0 mm
Base Resolution	128
Phase Resolution	100 %
Interpolation	Off

Resolution - Acceleration

Acceleration Mode	None
Phase Partial Fourier	Off
Asymmetric Echo	Off

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Distortion Correction	2D
Normalize	Off
Image Filter	Off

Geometry - Common

Slice Group	1
Slices	1
Distance Factor	20 %
Position	Isocenter
Orientation	Transversal
Phase Encoding Dir.	A >> P
Phase Oversampling	0 %
FOV Read	300 mm
FOV Phase	100.0 %
Slice Thickness	5.0 mm
TR	100.0 ms
Multi-Slice Mode	Interleaved
Series	Interleaved
Concatenations	1

Geometry - AutoAlign

Slice Group	1
Position	Isocenter
Orientation	Transversal
Phase Encoding Dir.	A >> P
AutoAlign	---
Initial Position	Isocenter
L	0.0 mm
P	0.0 mm
H	0.0 mm
Initial Orientation	Transversal
Initial Rotation	0.00 deg

Geometry - Saturation

Saturation Mode	Standard

Geometry - Saturation

Special Saturation	None

Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Table Position	0 mm
Table Position	H
Inline Composing	Off

System - Miscellaneous

Coil Selection	Auto Coil Select
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Adaptive Combine
Matrix Optimization	Off
Coil Focus	Flat

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Tune up
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	Isocenter
Orientation	Transversal
Rotation	0.00 deg
A >> P	263 mm
R >> L	350 mm
F >> H	350 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Slice-sel.
LR Balancing	Off

System - Tx/Rx

Frequency 1H	123.249442 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	100.0 ms
Segments	1
Concatenations	1

Physio - Cardiac

Fat-Water Contrast	Standard
Magn. Preparation	None
Dark Blood	Off
FOV Read	300 mm
FOV Phase	100.0 %
Phase Resolution	100 %

Physio - PACE

Resp. Control	Off
Concatenations	1

Inline - Liver

Liver Registration	Off
Save Original Images	On

Inline - Subtraction

Subtract	Off
Measurements	1
StdDev	Off
Save Original Images	On

Inline - MIP

MIP Sag	Off
MIP Cor	Off
MIP Tra	Off
MIP Time	Off
Radial MIP	Off
Save Original Images	On
MPR Sag	Off
MPR Cor	Off
MPR Tra	Off

Inline - Soft Tissue

Wash-in	Off
Wash-out	Off
TTP	Off
PEI	Off
MIP Time	Off
Measurements	1

Inline - Composing

Inline Composing	Off

Sequence - Part 1

Sequence Name	fl
Dimension	2D
Excitation	Slice-sel.
RF Pulse Type	Normal
Gradient Mode	Fast
Flow Compensation	None
Bandwidth	260 Hz/Px
Asymmetric Echo	Off
Segments	1

Sequence - Part 2

Introduction	On
RF Spoiling	On
Acoustic noise reduction	Off

Sequence - Special

MT Flip Angle	370 degrees
MT Offset	1500 Hz

Sequence - Assistant

SAR Assistant	Off
Allowed Delay	0 s

\\Research\Investigators\Frederick\allcustomer_20260909\can_neuromelanin_pk
TA: 14 sec Coil Selection: Auto Voxel Size: 2.3×2.3×5.0 mm³ Acc:: None Rel. SNR: 1.00

Properties

Start measurement without further preparation	On
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
Graphic segment	Default
Inline Movie	Off

Routine

Slice Group	1
Slices	1
Distance Factor	20 %
Position	Isocenter
Orientation	Transversal
Phase Encoding Dir.	A >> P
Phase Oversampling	0 %
FOV Read	300 mm
FOV Phase	100.0 %
Slice Thickness	5.0 mm
TR	100.0 ms
TE	10.00 ms
Averages	1
Concatenations	1
AutoAlign	---

Contrast - Common

TR	100.0 ms
TE	10.00 ms
MTC	Off
Magn. Preparation	None
Flip Angle	25 deg
Fat-Water Contrast	Standard
Dark Blood	Off
Contrasts	1
SWI	Off
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	1
Multiple Series	Each Measurement

Resolution - Common

FOV Read	300 mm
FOV Phase	100.0 %
Slice Thickness	5.0 mm
Base Resolution	128
Phase Resolution	100 %
Interpolation	Off

Resolution - Acceleration

Acceleration Mode	None
Phase Partial Fourier	Off
Asymmetric Echo	Off

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Distortion Correction	2D
Normalize	Off
Image Filter	Off

Geometry - Common

Slice Group	1
Slices	1
Distance Factor	20 %
Position	Isocenter
Orientation	Transversal
Phase Encoding Dir.	A >> P
Phase Oversampling	0 %
FOV Read	300 mm
FOV Phase	100.0 %
Slice Thickness	5.0 mm
TR	100.0 ms
Multi-Slice Mode	Interleaved
Series	Interleaved
Concatenations	1

Geometry - AutoAlign

Slice Group	1
Position	Isocenter
Orientation	Transversal
Phase Encoding Dir.	A >> P
AutoAlign	---
Initial Position	Isocenter
L	0.0 mm
P	0.0 mm
H	0.0 mm
Initial Orientation	Transversal
Initial Rotation	0.00 deg

Geometry - Saturation

Saturation Mode	Standard

Geometry - Saturation

Special Saturation	None

Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Table Position	0 mm
Table Position	H
Inline Composing	Off

System - Miscellaneous

Coil Selection	Auto Coil Select
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Adaptive Combine
Matrix Optimization	Off
Coil Focus	Flat

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Tune up
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	Isocenter
Orientation	Transversal
Rotation	0.00 deg
A >> P	263 mm
R >> L	350 mm
F >> H	350 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Slice-sel.
LR Balancing	Off

System - Tx/Rx

Frequency 1H	123.249442 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	100.0 ms
Segments	1
Concatenations	1

Physio - Cardiac

Fat-Water Contrast	Standard
Magn. Preparation	None
Dark Blood	Off
FOV Read	300 mm
FOV Phase	100.0 %
Phase Resolution	100 %

Physio - PACE

Resp. Control	Off
Concatenations	1

Inline - Liver

Liver Registration	Off
Save Original Images	On

Inline - Subtraction

Subtract	Off
Measurements	1
StdDev	Off
Save Original Images	On

Inline - MIP

MIP Sag	Off
MIP Cor	Off
MIP Tra	Off
MIP Time	Off
Radial MIP	Off
Save Original Images	On
MPR Sag	Off
MPR Cor	Off
MPR Tra	Off

Inline - Soft Tissue

Wash-in	Off
Wash-out	Off
TTP	Off
PEI	Off
MIP Time	Off
Measurements	1

Inline - Composing

Inline Composing	Off

Sequence - Part 1

Sequence Name	fl
Dimension	2D
Excitation	Slice-sel.
RF Pulse Type	Normal
Gradient Mode	Fast
Flow Compensation	None
Bandwidth	260 Hz/Px
Asymmetric Echo	Off
Segments	1

Sequence - Part 2

Introduction	On
RF Spoiling	On
Acoustic noise reduction	Off

Sequence - Special

MT Flip Angle	370 degrees
MT Offset	1500 Hz
Apply Partial Kspace Coverage	Off

Sequence - Assistant

SAR Assistant	Off
Allowed Delay	0 s

\\Research\Investigators\Frederick\allcustomer_20260909\cmrr_mbep2d_bold
TA: 8 sec Coil Selection: Auto Voxel Size: 3.9×3.9×5.0 mm³ Acc:: None Rel. SNR: 1.00

Properties

Start measurement without further preparation	On
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
Graphic segment	Default
Inline Movie	Off

Routine

Slice Group	1
Slices	1
Distance Factor	50 %
Position	Isocenter
Orientation	Transversal
Phase Encoding Dir.	A >> P
Phase Oversampling	0 %
FOV Read	500 mm
FOV Phase	100.0 %
Slice Thickness	5.0 mm
TR	250.0 ms
TE	60.00 ms
Averages	1
Multi-band accel. factor	1
AutoAlign	---

Contrast - Common

TR	250.0 ms
TE	60.00 ms
MTC	Off
Magn. Preparation	None
Flip Angle	90 deg
Fat-Water Contrast	Fat Saturation
Contrasts	1
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	20
Delay in TR	0.00 ms

Resolution - Common

FOV Read	500 mm
FOV Phase	100.0 %
Slice Thickness	5.0 mm
Base Resolution	128
Phase Resolution	100 %
Interpolation	Off

Resolution - Acceleration

Acceleration Mode	None
Phase Partial Fourier	Off

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Hamming	Off
Distortion Correction	2D
Static Field Correction	Off
Normalize	Off

Geometry - Common

Slice Group	1
Slices	1
Distance Factor	50 %
Position	Isocenter
Orientation	Transversal
Phase Encoding Dir.	A >> P
Phase Oversampling	0 %
FOV Read	500 mm
FOV Phase	100.0 %
Slice Thickness	5.0 mm
TR	250.0 ms
Multi-Slice Mode	Interleaved
Series	Interleaved
Multi-band accel. factor	1

Geometry - AutoAlign

Slice Group	1
Position	Isocenter
Orientation	Transversal
Phase Encoding Dir.	A >> P
AutoAlign	---
Initial Position	Isocenter
L	0.0 mm
P	0.0 mm
H	0.0 mm
Initial Orientation	Transversal
Initial Rotation	0.00 deg

Geometry - Saturation

Special Saturation	None

Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Table Position	0 mm
Table Position	H
Inline Composing	Off

System - Miscellaneous

Coil Selection	Auto Coil Select
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Sum of Squares
Matrix Optimization	Off
Coil Focus	Flat

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Standard
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	Isocenter
Orientation	Transversal
Rotation	0.00 deg
A >> P	500 mm
R >> L	500 mm
F >> H	5 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Standard

System - Tx/Rx

Frequency 1H	123.249442 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	250.0 ms
Log Signals	Off
Multi-band accel. factor	1

BOLD

GLM Statistics	On
Ignore Meas. at Start	0
Ignore After Transition	0

BOLD

Model Transition States	On
Temp. Highpass Filter	On
Threshold	4.00
Paradigm Size	20
Meas[1]	Baseline
Meas[2]	Baseline
Meas[3]	Baseline
Meas[4]	Baseline
Meas[5]	Baseline
Meas[6]	Baseline
Meas[7]	Baseline
Meas[8]	Baseline
Meas[9]	Baseline
Meas[10]	Baseline
Meas[11]	Active
Meas[12]	Active
Meas[13]	Active
Meas[14]	Active
Meas[15]	Active
Meas[16]	Active
Meas[17]	Active
Meas[18]	Active
Meas[19]	Active
Meas[20]	Active
Motion Correction	On
Spatial Filter	Off
Measurements	20
Delay in TR	0.00 ms

Sequence - Part 1

Sequence Name	epfid
Dimension	2D
Excitation	Standard
Gradient Mode	Fast
Flow Compensation	None
Bandwidth	1502 Hz/Px
Echo Spacing	0.71 ms
Free Echo Spacing	Off
EPI Factor	128

Sequence - Part 2

Introduction	Off
RF Spoiling	Off

Sequence - Special

Excite pulse duration	2560 us
Min. prep scans	0
Delay before PC scans	0 us
SENSE1 coil combine	Off
Invert RO/PE polarity	Off
Disable freq. update	Off
Suppress 16-bit DICOM	Off
Force equal slice timing	Off
FFT scale factor	1.00

Sequence - Special

Physio recording	Off
Triggering scheme	Standard

Sequence - Assistant

SAR Assistant	Off

\\Research\Investigators\Frederick\allcustomer_20260909\cmrr_mbep2d_diff
TA: 1 sec Coil Selection: Auto Voxel Size: 3.9×3.9×5.0 mm³ Acc:: None Rel. SNR: 1.00

Properties

Start measurement without further preparation	On
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
Graphic segment	Default
Inline Movie	Off

Routine

Slice Group	1
Slices	1
Distance Factor	50 %
Position	Isocenter
Orientation	Transversal
Phase Encoding Dir.	A >> P
Phase Oversampling	0 %
FOV Read	500 mm
FOV Phase	100.0 %
Slice Thickness	5.0 mm
TR	500.0 ms
TE	200.00 ms
Multi-band accel. factor	1
AutoAlign	---

Contrast - Common

TR	500.0 ms
TE	200.00 ms
MTC	Off
Magn. Preparation	None
Flip Angle	90 deg
Refocus flip angle	180 deg
Fat-Water Contrast	Fat Saturation
Grad. rev. fat suppr.	Enabled
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Multiple Series	Off
Delay in TR	0.00 ms

Resolution - Common

FOV Read	500 mm
FOV Phase	100.0 %

Resolution - Common

Slice Thickness	5.0 mm
Base Resolution	128
Phase Resolution	100 %
Interpolation	Off

Resolution - Acceleration

Acceleration Mode	None
Phase Partial Fourier	6/8

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Distortion Correction	Off
Static Field Correction	Off
Normalize	Off

Geometry - Common

Slice Group	1
Slices	1
Distance Factor	50 %
Position	Isocenter
Orientation	Transversal
Phase Encoding Dir.	A >> P
Phase Oversampling	0 %
FOV Read	500 mm
FOV Phase	100.0 %
Slice Thickness	5.0 mm
TR	500.0 ms
Multi-Slice Mode	Interleaved
Series	Interleaved
Multi-band accel. factor	1

Geometry - AutoAlign

Slice Group	1
Position	Isocenter
Orientation	Transversal
Phase Encoding Dir.	A >> P
AutoAlign	---
Initial Position	Isocenter
L	0.0 mm
P	0.0 mm
H	0.0 mm
Initial Orientation	Transversal
Initial Rotation	0.00 deg

Geometry - Navigator

Geometry - Saturation

Special Saturation	None

Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Table Position	0 mm
Table Position	H
Inline Composing	Off

System - Miscellaneous

Coil Selection	Auto Coil Select
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Sum of Squares
Matrix Optimization	Off
Coil Focus	Flat

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Standard
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	Isocenter
Orientation	Transversal
Rotation	0.00 deg
A >> P	500 mm
R >> L	500 mm
F >> H	5 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Standard

System - Tx/Rx

Frequency 1H	123.249442 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	500.0 ms
Multi-band accel. factor	1

Physio - PACE

Resp. Control	Off
Multi-band accel. factor	1

Diff

Diffusion Mode	MDDW
Diff. Directions	6
Diffusion Scheme	Bipolar
Diff. Weightings	1
b-value	0 s/mm²
Averages	1
Dynamic Field Correction	Off
Invert Gray Scale	Off
Diff. Weighted Images	On
Trace Weighted Images	Off
Tensor	Off
FA Maps	Off
ADC Maps	Off
Exponential ADC Maps	Off
Calculated Image	Off

Sequence - Part 1

Sequence Name	epse
Dimension	2D
Excitation	Standard
RF Pulse Type	Normal
Gradient Mode	Fast
Bandwidth	1502 Hz/Px
Echo Spacing	0.71 ms
Free Echo Spacing	Off
EPI Factor	128

Sequence - Part 2

Introduction	Off
RF Spoiling	Off
Phase Correction	Internal

Sequence - Special

Min. prep scans	0
Delay before PC scans	0 us
SENSE1 coil combine	Off
Invert RO/PE polarity	Off
PF omits higher k-space	Off
Force GPA balance	Off
Disable B1 control loop	Off
Disable freq. update	Off
Suppress 16-bit DICOM	Off
Force equal slice timing	Off
FFT scale factor	1.00
Physio recording	Off

Sequence - Assistant

SAR Assistant	Off

\\Research\Investigators\Frederick\allcustomer_20260909\cmrr_mbep2d_se

TA: 7 sec Coil Selection: Auto Voxel Size: 3.9×3.9×5.0 mm³ Acc:: None Rel. SNR: 1.00

Properties

Start measurement without further preparation	On
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
Graphic segment	Default
Inline Movie	Off

Routine

Slice Group	1
Slices	1
Distance Factor	50 %
Position	Isocenter
Orientation	Transversal
Phase Encoding Dir.	A >> P
Phase Oversampling	0 %
FOV Read	500 mm
FOV Phase	100.0 %
Slice Thickness	5.0 mm
TR	177.0 ms
TE	95.60 ms
Averages	1
Multi-band accel. factor	1
AutoAlign	---

Contrast - Common

TR	177.0 ms
TE	95.60 ms
MTC	Off
Magn. Preparation	None
Flip Angle	90 deg
Refocus flip angle	180 deg
Fat-Water Contrast	Fat Saturation
Grad. rev. fat suppr.	Enabled
Contrasts	1
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	20
Delay in TR	0.00 ms

Resolution - Common

FOV Read	500 mm
FOV Phase	100.0 %
Slice Thickness	5.0 mm
Base Resolution	128
Phase Resolution	100 %
Interpolation	Off

Resolution - Acceleration

Acceleration Mode	None
Phase Partial Fourier	Off

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Hamming	Off
Distortion Correction	2D
Static Field Correction	Off
Normalize	Off

Geometry - Common

Slice Group	1
Slices	1
Distance Factor	50 %
Position	Isocenter
Orientation	Transversal
Phase Encoding Dir.	A >> P
Phase Oversampling	0 %
FOV Read	500 mm
FOV Phase	100.0 %
Slice Thickness	5.0 mm
TR	177.0 ms
Multi-Slice Mode	Interleaved
Series	Interleaved
Multi-band accel. factor	1

Geometry - AutoAlign

Slice Group	1
Position	Isocenter
Orientation	Transversal
Phase Encoding Dir.	A >> P
AutoAlign	---
Initial Position	Isocenter
L	0.0 mm
P	0.0 mm
H	0.0 mm
Initial Orientation	Transversal
Initial Rotation	0.00 deg

Geometry - Saturation

Special Saturation	None

Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Table Position	0 mm
Table Position	H
Inline Composing	Off

System - Miscellaneous

Coil Selection	Auto Coil Select
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Sum of Squares
Matrix Optimization	Off
Coil Focus	Flat

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Standard
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	Isocenter
Orientation	Transversal
Rotation	0.00 deg
A >> P	500 mm
R >> L	500 mm
F >> H	5 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Standard

System - Tx/Rx

Frequency 1H	123.249442 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	177.0 ms
Log Signals	Off
Multi-band accel. factor	1

BOLD

GLM Statistics	On
Ignore Meas. at Start	0
Ignore After Transition	0

BOLD

Model Transition States	On
Temp. Highpass Filter	On
Threshold	4.00
Paradigm Size	20
Meas[1]	Baseline
Meas[2]	Baseline
Meas[3]	Baseline
Meas[4]	Baseline
Meas[5]	Baseline
Meas[6]	Baseline
Meas[7]	Baseline
Meas[8]	Baseline
Meas[9]	Baseline
Meas[10]	Baseline
Meas[11]	Active
Meas[12]	Active
Meas[13]	Active
Meas[14]	Active
Meas[15]	Active
Meas[16]	Active
Meas[17]	Active
Meas[18]	Active
Meas[19]	Active
Meas[20]	Active
Motion Correction	On
Spatial Filter	Off
Measurements	20
Delay in TR	0.00 ms

Sequence - Part 1

Sequence Name	epse
Dimension	2D
Excitation	Standard
RF Pulse Type	Normal
Gradient Mode	Fast
Bandwidth	1502 Hz/Px
Echo Spacing	0.71 ms
Free Echo Spacing	Off
EPI Factor	128

Sequence - Part 2

Introduction	Off

Sequence - Special

Min. prep scans	0
Delay before PC scans	0 us
SENSE1 coil combine	Off
Invert RO/PE polarity	Off
Force GPA balance	Off
Disable freq. update	Off
Suppress 16-bit DICOM	Off
Force equal slice timing	Off
FFT scale factor	1.00
Physio recording	Off

Sequence - Special

Triggering scheme	Standard

Sequence - Assistant

SAR Assistant	Off

\\Research\Investigators\Frederick\allcustomer_20260909\CMRR_XFL_mbPCASL
TA: 10:51 min Coil Selection: Auto Voxel Size: 2.5×2.5×2.3 mm³ Acc:: None Rel. SNR: 1.00

Properties

Start measurement without further preparation	On
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
Graphic segment	Default
Inline Movie	Off

Routine

Slice Group	1
Slices	60
Distance Factor	10 %
Position	Isocenter
Orientation	Transversal
Phase Encoding Dir.	A >> P
Phase Oversampling	0 %
FOV Read	215 mm
FOV Phase	100.0 %
Slice Thickness	2.3 mm
TR	7150.0 ms
TE	30.00 ms
Averages	1
Concatenations	1
AutoAlign	---

Contrast - Common

TR	7150.0 ms
TE	30.00 ms
MTC	Off
Flip Angle	90 deg
Fat-Water Contrast	Fat Saturation
Fat Saturation	Strong
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	90
Delay in TR	0.00 ms

Resolution - Common

FOV Read	215 mm
FOV Phase	100.0 %
Slice Thickness	2.3 mm

Resolution - Common

Base Resolution	86
Phase Resolution	100 %
Interpolation	Off

Resolution - Acceleration

Acceleration Mode	None
Phase Partial Fourier	6/8

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Hamming	Off
Distortion Correction	2D
Static Field Correction	Off
Normalize	Off

Geometry - Common

Slice Group	1
Slices	60
Distance Factor	10 %
Position	Isocenter
Orientation	Transversal
Phase Encoding Dir.	A >> P
Phase Oversampling	0 %
FOV Read	215 mm
FOV Phase	100.0 %
Slice Thickness	2.3 mm
TR	7150.0 ms
Multi-Slice Mode	Interleaved
Series	Ascending
Concatenations	1

Geometry - AutoAlign

Slice Group	1
Position	Isocenter
Orientation	Transversal
Phase Encoding Dir.	A >> P
AutoAlign	---
Initial Position	Isocenter
L	0.0 mm
P	0.0 mm
H	0.0 mm
Initial Orientation	Transversal
Initial Rotation	0.00 deg

Geometry - Saturation

Special Saturation	None

Geometry - Tim Planning Suite

Set-n-Go Protocol	Off

Geometry - Tim Planning Suite

Table Position	0 mm
Table Position	H
Inline Composing	Off

System - Miscellaneous

Coil Selection	Auto Coil Select
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Sum of Squares
Matrix Optimization	Off
Coil Focus	Flat

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Advanced
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	Isocenter
Orientation	Transversal
Rotation	0.00 deg
A >> P	215 mm
R >> L	215 mm
F >> H	150 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Standard

System - Tx/Rx

Frequency 1H	123.249442 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	7150.0 ms
Log Signals	Off
Concatenations	1

BOLD

GLM Statistics	On
Ignore Meas. at Start	0
Ignore After Transition	0
Model Transition States	On

BOLD

Temp. Highpass Filter	On
Threshold	4.00
Paradigm Size	20
Meas[1]	Baseline
Meas[2]	Baseline
Meas[3]	Baseline
Meas[4]	Baseline
Meas[5]	Baseline
Meas[6]	Baseline
Meas[7]	Baseline
Meas[8]	Baseline
Meas[9]	Baseline
Meas[10]	Baseline
Meas[11]	Active
Meas[12]	Active
Meas[13]	Active
Meas[14]	Active
Meas[15]	Active
Meas[16]	Active
Meas[17]	Active
Meas[18]	Active
Meas[19]	Active
Meas[20]	Active
Motion Correction	On
Spatial Filter	Off
Measurements	90
Delay in TR	0.00 ms

Sequence - Part 1

Sequence Name	epfid
Excitation	Standard
RF Pulse Type	Normal
Gradient Mode	Fast
Bandwidth	2326 Hz/Px
Echo Spacing	0.57 ms
Free Echo Spacing	Off
EPI Factor	86

Sequence - Part 2

Introduction	Off

Sequence - Special

PCASL Options	mdPCASL
Labeling Duration.	1500000 us.
Postlabeling delay.	1641860 us.

Sequence - Assistant

SAR Assistant	Off

\\Research\Investigators\Frederick\allcustomer_20260909\dkd_svs_sLASER
TA: 15 sec Coil Selection: Auto Vol: 20×20×20 mm³ Rel. SNR: 1.00

Properties

Start measurement without further preparation	On
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
Graphic segment	Default
Inline Movie	Off

Routine

Position	Isocenter
Orientation	Transversal
Rotation	0.00 deg
Vol A >> P	20 mm
Vol R >> L	20 mm
Vol F >> H	20 mm
TR	5000.0 ms
TE 1	8.00 ms
TE 2	12.00 ms
TE 3	10.00 ms
Averages	1

Contrast - Common

TR	5000.0 ms
TE 1	8.00 ms
TE 2	12.00 ms
TE 3	10.00 ms
Total TE	30.00 ms
Excitation Flip angle	90 deg
Refocusing Flip angle	180 deg
Preparation Scans	2
Averages	1
VAPOR WS	Off

Resolution - Common

Vector Size	2048
Normalize	Off

Geometry - Common

Position	Isocenter
Orientation	Transversal
Rotation	0.00 deg
Vol A >> P	20 mm
Vol R >> L	20 mm

Geometry - Common

Vol F >> H	20 mm

Geometry - AutoAlign

AutoAlign	---
Initial Position	Isocenter
L	0.0 mm
P	0.0 mm
H	0.0 mm
Initial Orientation	Transversal
Initial Rotation	0.00 deg

Geometry - Navigator

System - Miscellaneous

Coil Selection	Auto Coil Select
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Advanced
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	Isocenter
Orientation	Transversal
Rotation	0.00 deg
A >> P	20 mm
R >> L	20 mm
F >> H	20 mm
Reset	Off

System - pTx

B1 Shim	TrueForm

System - Tx/Rx

Frequency 1H	123.249442 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None

Physio - Signal

TR	5000.0 ms

Sequence - Common

Sequence Name	slaser
Preparation Scans	2
Delta Frequency	0.00 ppm
Phase Cycling	Auto
Save Uncombined	Off
Bandwidth	6000 Hz
Acquisition Duration	341 ms
Remove Oversampling	On

Sequence - Special

Excitation duration	2000.00 us
Refocusing duration	4500.00 us
AFP Type	GOIA-WURST
Bandwidth_1ms	45 kHz
HSn modulation	16
Gradient factor	85 %
AutoShim	Off
GOIA/FOCI	On
Inversion pulse	Off
FA AutoCalib	Off
AutoVOI	Off
Advanced User	Off
Calibration Type	None
Debug Type	None
Spinal Cord Navigator	Off
Gradient Max. Amplitude	37 mT/m
Ramp time	200 us

\\Research\Investigators\Frederick\allcustomer_20260909\eja_csi_fid
TA: 1:42:33 h Coil Selection: Auto Voxel Size: 12.5×12.5×25.0 mm³ Rel. SNR: 1.00

Properties

Start measurement without further preparation	On
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
Graphic segment	Default
Inline Movie	Off

Routine

Position	Isocenter
Orientation	Transversal
Rotation	0.00 deg
FOV A >> P	200 mm
FOV R >> L	200 mm
FOV F >> H	200 mm
TR	3000.0 ms
TE	5.00 ms
Averages	1

Contrast - Common

TR	3000.0 ms
TE	5.00 ms
Flip Angle	90 deg
Preparation Scans	0
Averages	1
Water Suppression	None

Resolution - Common

FOV A >> P	200 mm
FOV R >> L	200 mm
FOV F >> H	200 mm
Scan Res. A >> P	16
Scan Res. R >> L	16
Scan Res. F >> H	8
Interpol. Res. A >> P	16
Interpol. Res. R >> L	16
Interpol. Res. F >> H	8
Hamming	Off
Dimension	3D
Phase Encoding	Full
Vector Size	2048
Normalize	Off

Geometry - Common

Position	Isocenter
Orientation	Transversal
Rotation	0.00 deg
FOV A >> P	200 mm
FOV R >> L	200 mm
FOV F >> H	200 mm

Geometry - AutoAlign

Slab Group	1
Position	Isocenter
Orientation	Transversal
Phase Encoding Dir.	A >> P
Initial Position	Isocenter
L	0.0 mm
P	0.0 mm
H	0.0 mm
Initial Orientation	Transversal
Initial Rotation	0.00 deg

System - Miscellaneous

Coil Selection	Auto Coil Select
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Advanced
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off
Adj. Water Suppr.	Off

System - Adjust Volume

Position	Isocenter
Orientation	Transversal
Rotation	0.00 deg
A >> P	200 mm
R >> L	200 mm
F >> H	200 mm
Reset	Off

System - pTx

B1 Shim	TrueForm

System - Tx/Rx

Frequency 1H	123.249442 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	3000.0 ms

Sequence - Common

Sequence Name	fid
Preparation Scans	0
Delta Frequency	0.00 ppm
Measurements	1
Dimension	3D
Phase Cycling	None
Save Uncombined	Off
Bandwidth	2000 Hz
Acquisition Duration	1024 ms
Remove Oversampling	On

Sequence - Nuclei

TX/RX Nucleus	1H
TX/RX Delta Frequency	0 Hz
TX Nucleus	None
TX Delta Frequency	0 Hz

Sequence - Special

Excite pulse duration	1000 us
Acq. window shift	200 us
Min. settling delay	300 us
Gradient ramp time	140 us
Use expt. SAR calc	Off
Check RF clipping	Off
Send ref. scans	Off
Inversion pulse	Off
Shift RO frequency	Off
Debug loop type	None

\\Research\Investigators\Frederick\allcustomer_20260909\eja_csi_laser
TA: 13:09 min Coil Selection: Auto Voxel Size: 12.5×12.5×20.0 mm³ Rel. SNR: 1.00

Properties

Start measurement without further preparation	On
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
Graphic segment	Default
Inline Movie	Off

Routine

Position	Isocenter
Orientation	Transversal
Rotation	0.00 deg
FOV A >> P	200 mm
FOV R >> L	200 mm
Thickness F >> H	20 mm
Vol A >> P	20 mm
Vol R >> L	20 mm
TR	3000.0 ms
TE	135.00 ms
Averages	1

Contrast - Common

TR	3000.0 ms
TE	135.00 ms
Excite flip angle	90 deg
Refocus flip angle	180 deg
Preparation Scans	4
Averages	1
VAPOR	Water Saturation
VAPOR suppr.	Water Suppr.
Water s. BW	135.00 Hz
Water s. Delta Pos.	0.00 ppm

Resolution - Common

FOV A >> P	200 mm
FOV R >> L	200 mm
Thickness F >> H	20 mm
Scan Res. A >> P	16
Scan Res. R >> L	16
Interpol. Res. A >> P	16
Interpol. Res. R >> L	16
Hamming	Off
Dimension	2D
Phase Encoding	Full

Resolution - Common

Vector Size	2048
Normalize	Off

Geometry - Common

Position	Isocenter
Orientation	Transversal
Rotation	0.00 deg
FOV A >> P	200 mm
FOV R >> L	200 mm
Thickness F >> H	20 mm
Vol A >> P	20 mm
Vol R >> L	20 mm

Geometry - AutoAlign

Slice Group	1
Position	Isocenter
Orientation	Transversal
Phase Encoding Dir.	A >> P
AutoAlign	---
Initial Position	Isocenter
L	0.0 mm
P	0.0 mm
H	0.0 mm
Initial Orientation	Transversal
Initial Rotation	0.00 deg

System - Miscellaneous

Coil Selection	Auto Coil Select
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Advanced
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off
Adj. Water Suppr.	Off

System - Adjust Volume

Position	Isocenter
Orientation	Transversal
Rotation	0.00 deg
A >> P	20 mm
R >> L	20 mm

System - Adjust Volume

F >> H	20 mm
Reset	Off

System - pTx

B1 Shim	TrueForm

System - Tx/Rx

Frequency 1H	123.249442 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	3000.0 ms

Sequence - Common

Sequence Name	laser
Preparation Scans	4
Delta Frequency	-1.70 ppm
Measurements	1
Dimension	2D
Phase Cycling	Auto
Save Uncombined	Off
Bandwidth	2000 Hz
Acquisition Duration	1024 ms
Remove Oversampling	On

Sequence - Nuclei

TX/RX Nucleus	1H
TX/RX Delta Frequency	0 Hz
TX Nucleus	None
TX Delta Frequency	0 Hz

Sequence - Special

Excite pulse duration	8960 us
Refocus pulse duration	8960 us
Spoiler max. amplitude	20.00 mT/m
Spoiler amp. ratio	3.00
Spoiler duration	1300 us
Acq. window shift	200 us
Min. settling delay	300 us
Gradient ramp time	140 us
Spoiling scheme	0
HS refoc. pulse N	1
HS refoc. pulse R	20
VAPOR flip angle	80 deg
VAPOR delay 8	28 ms
VAPOR delay 7	74 ms
VAPOR delay 6	68 ms
VAPOR delay 5	106 ms
VAPOR delay 4	105 ms
VAPOR delay 3	146 ms
VAPOR delay 2	100 ms

Sequence - Special

VAPOR delay 1	150 ms
Use expt. SAR calc	Off
Check RF clipping	Off
Send ref. scans	Off
Inversion pulse	Off
GOIA refoc. pulses	Off
Rect excit. pulse	Off
Invert SS grad. pol.	Off
Shift RO frequency	Off
Debug loop type	None

\\Research\Investigators\Frederick\allcustomer_20260909\eja_csi_mslaser
TA: 25:57 min Coil Selection: Auto Voxel Size: 12.5×12.5×20.0 mm³ Rel. SNR: 1.00

Properties

Start measurement without further preparation	On
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
Graphic segment	Default
Inline Movie	Off

Routine

Position	Isocenter
Orientation	Transversal
Rotation	0.00 deg
FOV A >> P	200 mm
FOV R >> L	200 mm
Thickness F >> H	20 mm
Vol A >> P	20 mm
Vol R >> L	20 mm
TR	3000.0 ms
TE	135.00 ms
Averages	1

Contrast - Common

TR	3000.0 ms
TE	135.00 ms
Excite flip angle	90 deg
Refocus flip angle	180 deg
Preparation Scans	4
Averages	1
VAPOR	Water Saturation
VAPOR suppr.	Water Suppr.
Water s. BW	135.00 Hz
Water s. Delta Pos.	0.00 ppm

Resolution - Common

FOV A >> P	200 mm
FOV R >> L	200 mm
Thickness F >> H	20 mm
Scan Res. A >> P	16
Scan Res. R >> L	16
Interpol. Res. A >> P	16
Interpol. Res. R >> L	16
Hamming	Off
Dimension	2D
Phase Encoding	Full

Resolution - Common

Vector Size	2048
Normalize	Off

Geometry - Common

Position	Isocenter
Orientation	Transversal
Rotation	0.00 deg
FOV A >> P	200 mm
FOV R >> L	200 mm
Thickness F >> H	20 mm
Vol A >> P	20 mm
Vol R >> L	20 mm

Geometry - AutoAlign

Slice Group	1
Position	Isocenter
Orientation	Transversal
Phase Encoding Dir.	A >> P
AutoAlign	---
Initial Position	Isocenter
L	0.0 mm
P	0.0 mm
H	0.0 mm
Initial Orientation	Transversal
Initial Rotation	0.00 deg

System - Miscellaneous

Coil Selection	Auto Coil Select
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Advanced
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off
Adj. Water Suppr.	Off

System - Adjust Volume

Position	Isocenter
Orientation	Transversal
Rotation	0.00 deg
A >> P	20 mm
R >> L	20 mm

System - Adjust Volume

F >> H	20 mm
Reset	Off

System - pTx

B1 Shim	TrueForm

System - Tx/Rx

Frequency 1H	123.249442 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	3000.0 ms

Sequence - Common

Sequence Name	mslsr
Preparation Scans	4
Delta Frequency	-1.70 ppm
Measurements	1
Dimension	2D
Phase Cycling	Auto
Save Uncombined	Off
Bandwidth	2000 Hz
Acquisition Duration	1024 ms
Remove Oversampling	On

Sequence - Nuclei

TX/RX Nucleus	1H
TX/RX Delta Frequency	0 Hz
TX Nucleus	None
TX Delta Frequency	0 Hz

Sequence - Special

Excite pulse duration	8960 us
Refocus pulse duration	8960 us
Spoiler max. amplitude	20.00 mT/m
Spoiler amp. ratio	3.00
Refocus grad. factor	1.00
OVS slab thickness	80.00 mm
OVS slab pos. offset	5.00 mm
OVS delta frequency	0.00 ppm
Spoiler duration	1300 us
Acq. window shift	200 us
Min. settling delay	300 us
Gradient ramp time	140 us
MEGA flip angle	180 deg
HS refoc. pulse N	1
HS refoc. pulse R	20
VAPOR flip angle	80 deg
VAPOR delay 8	28 ms
VAPOR delay 7	74 ms
OVS pulse duration	5120 us

Sequence - Special

OVS HS pulse N	1
OVS HS pulse R	40
OVS flip angle RO	90 deg
OVS flip angle PH	0 deg
OVS flip angle SL	0 deg
VAPOR delay 6	68 ms
VAPOR delay 5	106 ms
VAPOR delay 4	105 ms
VAPOR delay 3	146 ms
VAPOR delay 2	100 ms
VAPOR delay 1	150 ms
Enable OVS	On
Use expt. SAR calc	Off
Check RF clipping	Off
Send ref. scans	Off
MEGA water suppr.	Off
Inversion pulse	Off
GOIA refoc. pulses	Off
Asym. excit. pulse	Off
Invert SS grad. pol.	Off
Shift RO frequency	Off
Debug loop type	None
Number of label freqs.	2
Editing pulse freq. [1]	9.00 ppm
Editing pulse freq. [2]	1.90 ppm
Editing pulse BW	100.00 Hz

\\Research\Investigators\Frederick\allcustomer_20260909\eja_csi_press
TA: 13:09 min Coil Selection: Auto Voxel Size: 12.5×12.5×20.0 mm³ Rel. SNR: 1.00

Properties

Start measurement without further preparation	On
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
Graphic segment	Default
Inline Movie	Off

Routine

Position	Isocenter
Orientation	Transversal
Rotation	0.00 deg
FOV A >> P	200 mm
FOV R >> L	200 mm
Thickness F >> H	20 mm
Vol A >> P	20 mm
Vol R >> L	20 mm
TR	3000.0 ms
TE	135.00 ms
Averages	1

Contrast - Common

TR	3000.0 ms
TE	135.00 ms
Excite flip angle	90 deg
Refocus flip angle	180 deg
Preparation Scans	4
Averages	1
VAPOR	Water Saturation
VAPOR suppr.	Water Suppr.
Water s. BW	135.00 Hz
Water s. Delta Pos.	0.00 ppm

Resolution - Common

FOV A >> P	200 mm
FOV R >> L	200 mm
Thickness F >> H	20 mm
Scan Res. A >> P	16
Scan Res. R >> L	16
Interpol. Res. A >> P	16
Interpol. Res. R >> L	16
Hamming	Off
Dimension	2D
Phase Encoding	Full

Resolution - Common

Vector Size	2048
Normalize	Off

Geometry - Common

Position	Isocenter
Orientation	Transversal
Rotation	0.00 deg
FOV A >> P	200 mm
FOV R >> L	200 mm
Thickness F >> H	20 mm
Vol A >> P	20 mm
Vol R >> L	20 mm

Geometry - AutoAlign

Slice Group	1
Position	Isocenter
Orientation	Transversal
Phase Encoding Dir.	A >> P
AutoAlign	---
Initial Position	Isocenter
L	0.0 mm
P	0.0 mm
H	0.0 mm
Initial Orientation	Transversal
Initial Rotation	0.00 deg

System - Miscellaneous

Coil Selection	Auto Coil Select
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Advanced
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off
Adj. Water Suppr.	Off

System - Adjust Volume

Position	Isocenter
Orientation	Transversal
Rotation	0.00 deg
A >> P	20 mm
R >> L	20 mm

System - Adjust Volume

F >> H	20 mm
Reset	Off

System - pTx

B1 Shim	TrueForm

System - Tx/Rx

Frequency 1H	123.249442 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	3000.0 ms

Sequence - Common

Sequence Name	press
Preparation Scans	4
Delta Frequency	-1.70 ppm
Measurements	1
Dimension	2D
Phase Cycling	Auto
Save Uncombined	Off
Bandwidth	2000 Hz
Acquisition Duration	1024 ms
Remove Oversampling	On

Sequence - Nuclei

TX/RX Nucleus	1H
TX/RX Delta Frequency	0 Hz
TX Nucleus	None
TX Delta Frequency	0 Hz

Sequence - Special

Excite pulse duration	2600 us
Refocus pulse duration	5200 us
Spoiler max. amplitude	20.00 mT/m
Spoiler amp. ratio	3.00
Refocus grad. factor	1.00
OVS slab thickness	80.00 mm
OVS slab pos. offset	5.00 mm
OVS delta frequency	0.00 ppm
Spoiler duration	1300 us
Acq. window shift	200 us
Min. settling delay	300 us
Gradient ramp time	140 us
Forced min. TE1	0 ms
VAPOR flip angle	80 deg
VAPOR delay 8	28 ms
VAPOR delay 7	74 ms
OVS pulse duration	5120 us
OVS HS pulse N	1
OVS HS pulse R	40

Sequence - Special

OVS flip angle RO	90 deg
OVS flip angle PH	90 deg
OVS flip angle SL	90 deg
VAPOR delay 6	68 ms
VAPOR delay 5	106 ms
VAPOR delay 4	105 ms
VAPOR delay 3	146 ms
VAPOR delay 2	100 ms
VAPOR delay 1	150 ms
Enable OVS	On
Use expt. SAR calc	Off
Check RF clipping	Off
Send ref. scans	Off
Inversion pulse	Off
'RR' refoc. pulse	Off
Invert SS grad. pol.	Off
Shift RO frequency	Off
Debug loop type	None

\\Research\Investigators\Frederick\allcustomer_20260909\eja_csi_slaser
TA: 13:09 min Coil Selection: Auto Voxel Size: 12.5×12.5×20.0 mm³ Rel. SNR: 1.00

Properties

Start measurement without further preparation	On
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
Graphic segment	Default
Inline Movie	Off

Routine

Position	Isocenter
Orientation	Transversal
Rotation	0.00 deg
FOV A >> P	200 mm
FOV R >> L	200 mm
Thickness F >> H	20 mm
Vol A >> P	20 mm
Vol R >> L	20 mm
TR	3000.0 ms
TE	135.00 ms
Averages	1

Contrast - Common

TR	3000.0 ms
TE	135.00 ms
Excite flip angle	90 deg
Refocus flip angle	180 deg
Preparation Scans	4
Averages	1
VAPOR	Water Saturation
VAPOR suppr.	Water Suppr.
Water s. BW	135.00 Hz
Water s. Delta Pos.	0.00 ppm

Resolution - Common

FOV A >> P	200 mm
FOV R >> L	200 mm
Thickness F >> H	20 mm
Scan Res. A >> P	16
Scan Res. R >> L	16
Interpol. Res. A >> P	16
Interpol. Res. R >> L	16
Hamming	Off
Dimension	2D
Phase Encoding	Full

Resolution - Common

Vector Size	2048
Normalize	Off

Geometry - Common

Position	Isocenter
Orientation	Transversal
Rotation	0.00 deg
FOV A >> P	200 mm
FOV R >> L	200 mm
Thickness F >> H	20 mm
Vol A >> P	20 mm
Vol R >> L	20 mm

Geometry - AutoAlign

Slice Group	1
Position	Isocenter
Orientation	Transversal
Phase Encoding Dir.	A >> P
AutoAlign	---
Initial Position	Isocenter
L	0.0 mm
P	0.0 mm
H	0.0 mm
Initial Orientation	Transversal
Initial Rotation	0.00 deg

System - Miscellaneous

Coil Selection	Auto Coil Select
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Advanced
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off
Adj. Water Suppr.	Off

System - Adjust Volume

Position	Isocenter
Orientation	Transversal
Rotation	0.00 deg
A >> P	20 mm
R >> L	20 mm

System - Adjust Volume

F >> H	20 mm
Reset	Off

System - pTx

B1 Shim	TrueForm

System - Tx/Rx

Frequency 1H	123.249442 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	3000.0 ms

Sequence - Common

Sequence Name	slasr
Preparation Scans	4
Delta Frequency	-1.70 ppm
Measurements	1
Dimension	2D
Phase Cycling	Auto
Save Uncombined	Off
Bandwidth	2000 Hz
Acquisition Duration	1024 ms
Remove Oversampling	On

Sequence - Nuclei

TX/RX Nucleus	1H
TX/RX Delta Frequency	0 Hz
TX Nucleus	None
TX Delta Frequency	0 Hz

Sequence - Special

Excite pulse duration	8960 us
Refocus pulse duration	8960 us
Spoiler max. amplitude	20.00 mT/m
Spoiler amp. ratio	3.00
Refocus grad. factor	1.00
OVS slab thickness	80.00 mm
OVS slab pos. offset	5.00 mm
OVS delta frequency	0.00 ppm
Spoiler duration	1300 us
Acq. window shift	200 us
Min. settling delay	300 us
Gradient ramp time	140 us
HS refoc. pulse N	1
HS refoc. pulse R	20
VAPOR flip angle	80 deg
VAPOR delay 8	28 ms
VAPOR delay 7	74 ms
OVS pulse duration	5120 us
OVS HS pulse N	1

Sequence - Special

OVS HS pulse R	40
OVS flip angle RO	90 deg
OVS flip angle PH	0 deg
OVS flip angle SL	0 deg
VAPOR delay 6	68 ms
VAPOR delay 5	106 ms
VAPOR delay 4	105 ms
VAPOR delay 3	146 ms
VAPOR delay 2	100 ms
VAPOR delay 1	150 ms
Enable OVS	On
Use expt. SAR calc	Off
Check RF clipping	Off
Send ref. scans	Off
Inversion pulse	Off
GOIA refoc. pulses	Off
Asym. excit. pulse	Off
Invert SS grad. pol.	Off
Shift RO frequency	Off
Debug loop type	None

\\Research\Investigators\Frederick\allcustomer_20260909\eja_fid
TA: 12 sec Coil Selection: Auto Rel. SNR: 1.00

Properties

Start measurement without further preparation	On
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
Graphic segment	Default
Inline Movie	Off

Routine

TR	3000.0 ms
TE	5.00 ms
Averages	1

Contrast - Common

TR	3000.0 ms
TE	5.00 ms
Flip Angle	90 deg
Preparation Scans	0
Averages	1
Water Suppression	None

Resolution - Common

Vector Size	2048
Normalize	Off

Geometry - AutoAlign

Initial Position	Isocenter
L	0.0 mm
P	0.0 mm
H	0.0 mm
Initial Orientation	Transversal
Initial Rotation	0.00 deg

Geometry - Saturation

System - Miscellaneous

Coil Selection	Auto Coil Select
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Advanced
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off
Adj. Water Suppr.	Off

System - Adjust Volume

Position	Isocenter
Orientation	Transversal
Rotation	0.00 deg
A >> P	500 mm
R >> L	500 mm
F >> H	500 mm
Reset	Off

System - pTx

B1 Shim	TrueForm

System - Tx/Rx

Frequency 1H	123.249442 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	3000.0 ms

Sequence - Common

Sequence Name	fid
Preparation Scans	0
Delta Frequency	0.00 ppm
Measurements	1
Phase Cycling	None
Save Uncombined	Off
Bandwidth	2000 Hz
Acquisition Duration	1024 ms
Remove Oversampling	On

Sequence - Nuclei

TX/RX Nucleus	1H
TX/RX Delta Frequency	0 Hz
TX Nucleus	None
TX Delta Frequency	0 Hz

Sequence - Special

Excite pulse duration	1000 us

Sequence - Special

Acq. window shift	200 us
Min. settling delay	300 us
Gradient ramp time	140 us
Use expt. SAR calc	Off
Check RF clipping	Off
Send ref. scans	Off
Inversion pulse	Off
Shift RO frequency	Off
Debug loop type	None

\\Research\Investigators\Frederick\allcustomer_20260909\eja_svs_laser
TA: 3:39 min Coil Selection: Auto VoI: 20×20×20 mm³ Rel. SNR: 1.00

Properties

Start measurement without further preparation	On
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
Graphic segment	Default
Inline Movie	Off

Routine

Position	Isocenter
Orientation	Transversal
Rotation	0.00 deg
VoI A >> P	20 mm
VoI R >> L	20 mm
VoI F >> H	20 mm
TR	3000.0 ms
TE	135.00 ms
Averages	64

Contrast - Common

TR	3000.0 ms
TE	135.00 ms
Excite flip angle	90 deg
Refocus flip angle	180 deg
Preparation Scans	4
Averages	64
VAPOR	Water Saturation
VAPOR suppr.	Water Suppr.
Water s. BW	135.00 Hz
Water s. Delta Pos.	0.00 ppm

Resolution - Common

Vector Size	2048
Normalize	Off

Geometry - Common

Position	Isocenter
Orientation	Transversal
Rotation	0.00 deg
VoI A >> P	20 mm
VoI R >> L	20 mm
VoI F >> H	20 mm

Geometry - AutoAlign

AutoAlign	---
Initial Position	Isocenter
L	0.0 mm
P	0.0 mm
H	0.0 mm
Initial Orientation	Transversal
Initial Rotation	0.00 deg

System - Miscellaneous

Coil Selection	Auto Coil Select
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Advanced
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off
Adj. Water Suppr.	Off

System - Adjust Volume

Position	Isocenter
Orientation	Transversal
Rotation	0.00 deg
A >> P	20 mm
R >> L	20 mm
F >> H	20 mm
Reset	Off

System - pTx

B1 Shim	TrueForm

System - Tx/Rx

Frequency 1H	123.249442 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	3000.0 ms

Sequence - Common

Sequence Name	laser

Sequence - Common

Preparation Scans	4
Delta Frequency	-1.70 ppm
Measurements	1
Phase Cycling	Auto
Save Uncombined	Off
Bandwidth	2000 Hz
Acquisition Duration	1024 ms
Remove Oversampling	On

Sequence - Nuclei

TX/RX Nucleus	1H
TX/RX Delta Frequency	0 Hz
TX Nucleus	None
TX Delta Frequency	0 Hz

Sequence - Special

Excite pulse duration	8960 us
Refocus pulse duration	8960 us
Spoiler max. amplitude	20.00 mT/m
Spoiler amp. ratio	3.00
Spoiler duration	1300 us
Acq. window shift	200 us
Min. settling delay	300 us
Gradient ramp time	140 us
Spoiling scheme	0
HS refoc. pulse N	1
HS refoc. pulse R	20
VAPOR flip angle	80 deg
VAPOR delay 8	28 ms
VAPOR delay 7	74 ms
VAPOR delay 6	68 ms
VAPOR delay 5	106 ms
VAPOR delay 4	105 ms
VAPOR delay 3	146 ms
VAPOR delay 2	100 ms
VAPOR delay 1	150 ms
Resolve averages	Off
Use expt. SAR calc	Off
Check RF clipping	Off
Send ref. scans	Off
Inversion pulse	Off
GOIA refoc. pulses	Off
Rect excit. pulse	Off
Invert SS grad. pol.	Off
Shift RO frequency	Off
Debug loop type	None

\\Research\Investigators\Frederick\allcustomer_20260909\eja_svs_mslaser
TA: 6:51 min Coil Selection: Auto VoI: 20×20×20 mm³ Rel. SNR: 1.00

Properties

Start measurement without further preparation	On
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
Graphic segment	Default
Inline Movie	Off

Routine

Position	Isocenter
Orientation	Transversal
Rotation	0.00 deg
VoI A >> P	20 mm
VoI R >> L	20 mm
VoI F >> H	20 mm
TR	3000.0 ms
TE	135.00 ms
Averages	64

Contrast - Common

TR	3000.0 ms
TE	135.00 ms
Excite flip angle	90 deg
Refocus flip angle	180 deg
Preparation Scans	4
Averages	64
VAPOR	Water Saturation
VAPOR suppr.	Water Suppr.
Water s. BW	135.00 Hz
Water s. Delta Pos.	0.00 ppm

Resolution - Common

Vector Size	2048
Normalize	Off

Geometry - Common

Position	Isocenter
Orientation	Transversal
Rotation	0.00 deg
VoI A >> P	20 mm
VoI R >> L	20 mm
VoI F >> H	20 mm

Geometry - AutoAlign

AutoAlign	---
Initial Position	Isocenter
L	0.0 mm
P	0.0 mm
H	0.0 mm
Initial Orientation	Transversal
Initial Rotation	0.00 deg

System - Miscellaneous

Coil Selection	Auto Coil Select
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Advanced
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off
Adj. Water Suppr.	Off

System - Adjust Volume

Position	Isocenter
Orientation	Transversal
Rotation	0.00 deg
A >> P	20 mm
R >> L	20 mm
F >> H	20 mm
Reset	Off

System - pTx

B1 Shim	TrueForm

System - Tx/Rx

Frequency 1H	123.249442 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	3000.0 ms

Sequence - Common

Sequence Name	mslsr

Sequence - Common

Preparation Scans	4
Delta Frequency	-1.70 ppm
Measurements	1
Phase Cycling	Auto
Save Uncombined	Off
Bandwidth	2000 Hz
Acquisition Duration	1024 ms
Remove Oversampling	On

Sequence - Nuclei

TX/RX Nucleus	1H
TX/RX Delta Frequency	0 Hz
TX Nucleus	None
TX Delta Frequency	0 Hz

Sequence - Special

Excite pulse duration	8960 us
Refocus pulse duration	8960 us
Spoiler max. amplitude	20.00 mT/m
Spoiler amp. ratio	3.00
Refocus grad. factor	1.00
OVS slab thickness	80.00 mm
OVS slab pos. offset	5.00 mm
OVS delta frequency	0.00 ppm
Spoiler duration	1300 us
Acq. window shift	200 us
Min. settling delay	300 us
Gradient ramp time	140 us
MEGA flip angle	180 deg
HS refoc. pulse N	1
HS refoc. pulse R	20
VAPOR flip angle	80 deg
VAPOR delay 8	28 ms
VAPOR delay 7	74 ms
OVS pulse duration	5120 us
OVS HS pulse N	1
OVS HS pulse R	40
OVS flip angle RO	90 deg
OVS flip angle PH	0 deg
OVS flip angle SL	0 deg
VAPOR delay 6	68 ms
VAPOR delay 5	106 ms
VAPOR delay 4	105 ms
VAPOR delay 3	146 ms
VAPOR delay 2	100 ms
VAPOR delay 1	150 ms
Enable OVS	On
Resolve averages	Off
Use expt. SAR calc	Off
Check RF clipping	Off
Send ref. scans	Off
MEGA water suppr.	Off
Inversion pulse	Off
GOIA refoc. pulses	Off

Sequence - Special

Asym. excit. pulse	Off
Invert SS grad. pol.	Off
Shift RO frequency	Off
Debug loop type	None
Number of label freqs.	2
Editing pulse freq. [1]	9.00 ppm
Editing pulse freq. [2]	1.90 ppm
Editing pulse BW	100.00 Hz

\\Research\Investigators\Frederick\allcustomer_20260909\eja_svs_mpress
TA: 6:51 min Coil Selection: Auto Vol: 20×20×20 mm³ Rel. SNR: 1.00

Properties

Start measurement without further preparation	On
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
Graphic segment	Default
Inline Movie	Off

Routine

Position	Isocenter
Orientation	Transversal
Rotation	0.00 deg
Vol A >> P	20 mm
Vol R >> L	20 mm
Vol F >> H	20 mm
TR	3000.0 ms
TE	135.00 ms
Averages	64

Contrast - Common

TR	3000.0 ms
TE	135.00 ms
Excite flip angle	90 deg
Refocus flip angle	180 deg
Preparation Scans	4
Averages	64
VAPOR	Water Saturation
VAPOR suppr.	Water Suppr.
Water s. BW	135.00 Hz
Water s. Delta Pos.	0.00 ppm

Resolution - Common

Vector Size	2048
Normalize	Off

Geometry - Common

Position	Isocenter
Orientation	Transversal
Rotation	0.00 deg
Vol A >> P	20 mm
Vol R >> L	20 mm
Vol F >> H	20 mm

Geometry - AutoAlign

AutoAlign	---
Initial Position	Isocenter
L	0.0 mm
P	0.0 mm
H	0.0 mm
Initial Orientation	Transversal
Initial Rotation	0.00 deg

System - Miscellaneous

Coil Selection	Auto Coil Select
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Advanced
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off
Adj. Water Suppr.	Off

System - Adjust Volume

Position	Isocenter
Orientation	Transversal
Rotation	0.00 deg
A >> P	20 mm
R >> L	20 mm
F >> H	20 mm
Reset	Off

System - pTx

B1 Shim	TrueForm

System - Tx/Rx

Frequency 1H	123.249442 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	3000.0 ms

Sequence - Common

Sequence Name	mpres

Sequence - Common

Preparation Scans	4
Delta Frequency	-1.70 ppm
Measurements	1
Phase Cycling	Auto
Save Uncombined	Off
Bandwidth	2000 Hz
Acquisition Duration	1024 ms
Remove Oversampling	On

Sequence - Nuclei

TX/RX Nucleus	1H
TX/RX Delta Frequency	0 Hz
TX Nucleus	None
TX Delta Frequency	0 Hz

Sequence - Special

Excite pulse duration	2600 us
Refocus pulse duration	5200 us
Spoiler max. amplitude	20.00 mT/m
Spoiler amp. ratio	1.00
Refocus grad. factor	1.00
OVS slab thickness	80.00 mm
OVS slab pos. offset	5.00 mm
OVS delta frequency	0.00 ppm
Spoiler duration	1300 us
Acq. window shift	200 us
Min. settling delay	300 us
Gradient ramp time	140 us
Forced min. TE1	0 ms
MEGA flip angle	180 deg
VAPOR flip angle	80 deg
VAPOR delay 8	28 ms
VAPOR delay 7	74 ms
OVS pulse duration	5120 us
OVS HS pulse N	1
OVS HS pulse R	40
OVS flip angle RO	90 deg
OVS flip angle PH	90 deg
OVS flip angle SL	90 deg
VAPOR delay 6	68 ms
VAPOR delay 5	106 ms
VAPOR delay 4	105 ms
VAPOR delay 3	146 ms
VAPOR delay 2	100 ms
VAPOR delay 1	150 ms
Enable OVS	On
Resolve averages	Off
Use expt. SAR calc	Off
Check RF clipping	Off
Send ref. scans	Off
PRESS+4	Off
Symm. MEGA timing	On
MEGA water suppr.	Off
Inversion pulse	Off

Sequence - Special

'RR' refoc. pulse	Off
Invert SS grad. pol.	Off
Shift RO frequency	Off
Debug loop type	None
Number of label freqs.	2
Editing pulse freq. [1]	9.00 ppm
Editing pulse freq. [2]	1.90 ppm
Editing pulse BW	100.00 Hz

\\Research\Investigators\Frederick\allcustomer_20260909\eja_svs_press
TA: 3:39 min Coil Selection: Auto VoI: 20×20×20 mm³ Rel. SNR: 1.00

Properties

Start measurement without further preparation	On
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
Graphic segment	Default
Inline Movie	Off

Routine

Position	Isocenter
Orientation	Transversal
Rotation	0.00 deg
VoI A >> P	20 mm
VoI R >> L	20 mm
VoI F >> H	20 mm
TR	3000.0 ms
TE	135.00 ms
Averages	64

Contrast - Common

TR	3000.0 ms
TE	135.00 ms
Excite flip angle	90 deg
Refocus flip angle	180 deg
Preparation Scans	4
Averages	64
VAPOR	Water Saturation
VAPOR suppr.	Water Suppr.
Water s. BW	135.00 Hz
Water s. Delta Pos.	0.00 ppm

Resolution - Common

Vector Size	2048
Normalize	Off

Geometry - Common

Position	Isocenter
Orientation	Transversal
Rotation	0.00 deg
VoI A >> P	20 mm
VoI R >> L	20 mm
VoI F >> H	20 mm

Geometry - AutoAlign

AutoAlign	---
Initial Position	Isocenter
L	0.0 mm
P	0.0 mm
H	0.0 mm
Initial Orientation	Transversal
Initial Rotation	0.00 deg

System - Miscellaneous

Coil Selection	Auto Coil Select
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Advanced
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off
Adj. Water Suppr.	Off

System - Adjust Volume

Position	Isocenter
Orientation	Transversal
Rotation	0.00 deg
A >> P	20 mm
R >> L	20 mm
F >> H	20 mm
Reset	Off

System - pTx

B1 Shim	TrueForm

System - Tx/Rx

Frequency 1H	123.249442 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	3000.0 ms

Sequence - Common

Sequence Name	press

Sequence - Common

Preparation Scans	4
Delta Frequency	-1.70 ppm
Measurements	1
Phase Cycling	Auto
Save Uncombined	Off
Bandwidth	2000 Hz
Acquisition Duration	1024 ms
Remove Oversampling	On

Sequence - Nuclei

TX/RX Nucleus	1H
TX/RX Delta Frequency	0 Hz
TX Nucleus	None
TX Delta Frequency	0 Hz

Sequence - Special

Excite pulse duration	2600 us
Refocus pulse duration	5200 us
Spoiler max. amplitude	20.00 mT/m
Spoiler amp. ratio	3.00
Refocus grad. factor	1.00
OVS slab thickness	80.00 mm
OVS slab pos. offset	5.00 mm
OVS delta frequency	0.00 ppm
Spoiler duration	1300 us
Acq. window shift	200 us
Min. settling delay	300 us
Gradient ramp time	140 us
Forced min. TE1	0 ms
VAPOR flip angle	80 deg
VAPOR delay 8	28 ms
VAPOR delay 7	74 ms
OVS pulse duration	5120 us
OVS HS pulse N	1
OVS HS pulse R	40
OVS flip angle RO	90 deg
OVS flip angle PH	90 deg
OVS flip angle SL	90 deg
VAPOR delay 6	68 ms
VAPOR delay 5	106 ms
VAPOR delay 4	105 ms
VAPOR delay 3	146 ms
VAPOR delay 2	100 ms
VAPOR delay 1	150 ms
Enable OVS	On
Resolve averages	Off
Use expt. SAR calc	Off
Check RF clipping	Off
Send ref. scans	Off
Inversion pulse	Off
'RR' refoc. pulse	Off
Invert SS grad. pol.	Off
Shift RO frequency	Off
Debug loop type	None

\\Research\Investigators\Frederick\allcustomer_20260909\eja_svs_press_diff
TA: 3:39 min Coil Selection: Auto Vol: 20×20×20 mm³ Rel. SNR: 1.00

Properties

Start measurement without further preparation	On
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
Graphic segment	Default
Inline Movie	Off

Routine

Position	Isocenter
Orientation	Transversal
Rotation	0.00 deg
Vol A >> P	20 mm
Vol R >> L	20 mm
Vol F >> H	20 mm
TR	3000.0 ms
TE	135.00 ms
Averages	64

Contrast - Common

TR	3000.0 ms
TE	135.00 ms
Excite flip angle	90 deg
Refocus flip angle	180 deg
Preparation Scans	4
Averages	64
VAPOR	Water Saturation
VAPOR suppr.	Water Suppr.
Water s. BW	135.00 Hz
Water s. Delta Pos.	0.00 ppm

Resolution - Common

Vector Size	2048
Normalize	Off

Geometry - Common

Position	Isocenter
Orientation	Transversal
Rotation	0.00 deg
Vol A >> P	20 mm
Vol R >> L	20 mm
Vol F >> H	20 mm

Geometry - AutoAlign

AutoAlign	---
Initial Position	Isocenter
L	0.0 mm
P	0.0 mm
H	0.0 mm
Initial Orientation	Transversal
Initial Rotation	0.00 deg

System - Miscellaneous

Coil Selection	Auto Coil Select
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Advanced
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off
Adj. Water Suppr.	Off

System - Adjust Volume

Position	Isocenter
Orientation	Transversal
Rotation	0.00 deg
A >> P	20 mm
R >> L	20 mm
F >> H	20 mm
Reset	Off

System - pTx

B1 Shim	TrueForm

System - Tx/Rx

Frequency 1H	123.249442 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	3000.0 ms

Sequence - Common

Sequence Name	press

Sequence - Common

Preparation Scans	4
Delta Frequency	-1.70 ppm
Measurements	1
Phase Cycling	Auto
Save Uncombined	Off
Bandwidth	2000 Hz
Acquisition Duration	1024 ms
Remove Oversampling	On

Sequence - Nuclei

TX/RX Nucleus	1H
TX/RX Delta Frequency	0 Hz
TX Nucleus	None
TX Delta Frequency	0 Hz

Sequence - Special

Excite pulse duration	2600 us
Refocus pulse duration	5200 us
Spoiler max. amplitude	20.00 mT/m
Spoiler amp. ratio	3.00
Refocus grad. factor	1.00
OVS slab thickness	80.00 mm
OVS slab pos. offset	5.00 mm
OVS delta frequency	0.00 ppm
Spoiler duration	1300 us
Acq. window shift	200 us
Min. settling delay	300 us
Gradient ramp time	140 us
Forced min. TE1	0 ms
VAPOR flip angle	80 deg
VAPOR delay 8	28 ms
VAPOR delay 7	74 ms
OVS pulse duration	5120 us
OVS HS pulse N	1
OVS HS pulse R	40
OVS flip angle RO	90 deg
OVS flip angle PH	90 deg
OVS flip angle SL	90 deg
VAPOR delay 6	68 ms
VAPOR delay 5	106 ms
VAPOR delay 4	105 ms
VAPOR delay 3	146 ms
VAPOR delay 2	100 ms
VAPOR delay 1	150 ms
Diffusion weighting	Off
Enable OVS	On
Resolve averages	Off
Use expt. SAR calc	Off
Check RF clipping	Off
Send ref. scans	Off
Inversion pulse	Off
'RR' refoc. pulse	Off
Invert SS grad. pol.	Off
Shift RO frequency	Off

Sequence - Special

Debug loop type	None

\\Research\Investigators\Frederick\allcustomer_20260909\eja_svs_slaser
TA: 3:39 min Coil Selection: Auto VoI: 20×20×20 mm³ Rel. SNR: 1.00

Properties

Start measurement without further preparation	On
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
Graphic segment	Default
Inline Movie	Off

Routine

Position	Isocenter
Orientation	Transversal
Rotation	0.00 deg
VoI A >> P	20 mm
VoI R >> L	20 mm
VoI F >> H	20 mm
TR	3000.0 ms
TE	135.00 ms
Averages	64

Contrast - Common

TR	3000.0 ms
TE	135.00 ms
Excite flip angle	90 deg
Refocus flip angle	180 deg
Preparation Scans	4
Averages	64
VAPOR	Water Saturation
VAPOR suppr.	Water Suppr.
Water s. BW	135.00 Hz
Water s. Delta Pos.	0.00 ppm

Resolution - Common

Vector Size	2048
Normalize	Off

Geometry - Common

Position	Isocenter
Orientation	Transversal
Rotation	0.00 deg
VoI A >> P	20 mm
VoI R >> L	20 mm
VoI F >> H	20 mm

Geometry - AutoAlign

AutoAlign	---
Initial Position	Isocenter
L	0.0 mm
P	0.0 mm
H	0.0 mm
Initial Orientation	Transversal
Initial Rotation	0.00 deg

System - Miscellaneous

Coil Selection	Auto Coil Select
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Advanced
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off
Adj. Water Suppr.	Off

System - Adjust Volume

Position	Isocenter
Orientation	Transversal
Rotation	0.00 deg
A >> P	20 mm
R >> L	20 mm
F >> H	20 mm
Reset	Off

System - pTx

B1 Shim	TrueForm

System - Tx/Rx

Frequency 1H	123.249442 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	3000.0 ms

Sequence - Common

Sequence Name	slasr

Sequence - Common

Preparation Scans	4
Delta Frequency	-1.70 ppm
Measurements	1
Phase Cycling	Auto
Save Uncombined	Off
Bandwidth	2000 Hz
Acquisition Duration	1024 ms
Remove Oversampling	On

Sequence - Nuclei

TX/RX Nucleus	1H
TX/RX Delta Frequency	0 Hz
TX Nucleus	None
TX Delta Frequency	0 Hz

Sequence - Special

Excite pulse duration	8960 us
Refocus pulse duration	8960 us
Spoiler max. amplitude	20.00 mT/m
Spoiler amp. ratio	3.00
Refocus grad. factor	1.00
OVS slab thickness	80.00 mm
OVS slab pos. offset	5.00 mm
OVS delta frequency	0.00 ppm
Spoiler duration	1300 us
Acq. window shift	200 us
Min. settling delay	300 us
Gradient ramp time	140 us
HS refoc. pulse N	1
HS refoc. pulse R	20
VAPOR flip angle	80 deg
VAPOR delay 8	28 ms
VAPOR delay 7	74 ms
OVS pulse duration	5120 us
OVS HS pulse N	1
OVS HS pulse R	40
OVS flip angle RO	90 deg
OVS flip angle PH	0 deg
OVS flip angle SL	0 deg
VAPOR delay 6	68 ms
VAPOR delay 5	106 ms
VAPOR delay 4	105 ms
VAPOR delay 3	146 ms
VAPOR delay 2	100 ms
VAPOR delay 1	150 ms
Enable OVS	On
Resolve averages	Off
Use expt. SAR calc	Off
Check RF clipping	Off
Send ref. scans	Off
Inversion pulse	Off
GOIA refoc. pulses	Off
Asym. excit. pulse	Off
Invert SS grad. pol.	Off

Sequence - Special

Shift RO frequency	Off
Debug loop type	None

\\Research\Investigators\Frederick\allcustomer_20260909\eja_svs_slaser_diff
TA: 3:39 min Coil Selection: Auto VoI: 20×20×20 mm³ Rel. SNR: 1.00

Properties

Start measurement without further preparation	On
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
Graphic segment	Default
Inline Movie	Off

Routine

Position	Isocenter
Orientation	Transversal
Rotation	0.00 deg
VoI A >> P	20 mm
VoI R >> L	20 mm
VoI F >> H	20 mm
TR	3000.0 ms
TE	135.00 ms
Averages	64

Contrast - Common

TR	3000.0 ms
TE	135.00 ms
Excite flip angle	90 deg
Refocus flip angle	180 deg
Preparation Scans	4
Averages	64
VAPOR	Water Saturation
VAPOR suppr.	Water Suppr.
Water s. BW	135.00 Hz
Water s. Delta Pos.	0.00 ppm

Resolution - Common

Vector Size	2048
Normalize	Off

Geometry - Common

Position	Isocenter
Orientation	Transversal
Rotation	0.00 deg
VoI A >> P	20 mm
VoI R >> L	20 mm
VoI F >> H	20 mm

Geometry - AutoAlign

AutoAlign	---
Initial Position	Isocenter
L	0.0 mm
P	0.0 mm
H	0.0 mm
Initial Orientation	Transversal
Initial Rotation	0.00 deg

System - Miscellaneous

Coil Selection	Auto Coil Select
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Advanced
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off
Adj. Water Suppr.	Off

System - Adjust Volume

Position	Isocenter
Orientation	Transversal
Rotation	0.00 deg
A >> P	20 mm
R >> L	20 mm
F >> H	20 mm
Reset	Off

System - pTx

B1 Shim	TrueForm

System - Tx/Rx

Frequency 1H	123.249442 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	3000.0 ms

Sequence - Common

Sequence Name	slasr

Sequence - Common

Preparation Scans	4
Delta Frequency	-1.70 ppm
Measurements	1
Phase Cycling	Auto
Save Uncombined	Off
Bandwidth	2000 Hz
Acquisition Duration	1024 ms
Remove Oversampling	On

Sequence - Nuclei

TX/RX Nucleus	1H
TX/RX Delta Frequency	0 Hz
TX Nucleus	None
TX Delta Frequency	0 Hz

Sequence - Special

Excite pulse duration	8960 us
Refocus pulse duration	8960 us
Spoiler max. amplitude	20.00 mT/m
Spoiler amp. ratio	3.00
Refocus grad. factor	1.00
OVS slab thickness	80.00 mm
OVS slab pos. offset	5.00 mm
OVS delta frequency	0.00 ppm
Spoiler duration	1300 us
Acq. window shift	200 us
Min. settling delay	300 us
Gradient ramp time	140 us
HS refoc. pulse N	1
HS refoc. pulse R	20
VAPOR flip angle	80 deg
VAPOR delay 8	28 ms
VAPOR delay 7	74 ms
OVS pulse duration	5120 us
OVS HS pulse N	1
OVS HS pulse R	40
OVS flip angle RO	90 deg
OVS flip angle PH	0 deg
OVS flip angle SL	0 deg
VAPOR delay 6	68 ms
VAPOR delay 5	106 ms
VAPOR delay 4	105 ms
VAPOR delay 3	146 ms
VAPOR delay 2	100 ms
VAPOR delay 1	150 ms
Diffusion weighting	Off
Enable OVS	On
Resolve averages	Off
Use expt. SAR calc	Off
Check RF clipping	Off
Send ref. scans	Off
Inversion pulse	Off
GOIA refoc. pulses	Off
Asym. excit. pulse	Off

Sequence - Special

Invert SS grad. pol.	Off
Shift RO frequency	Off
Debug loop type	None

\\Research\Investigators\Frederick\allcustomer_20260909\eja_svs_steam
TA: 3:39 min Coil Selection: Auto VoI: 20×20×20 mm³ Rel. SNR: 1.00

Properties

Start measurement without further preparation	On
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
Graphic segment	Default
Inline Movie	Off

Routine

Position	Isocenter
Orientation	Transversal
Rotation	0.00 deg
VoI A >> P	20 mm
VoI R >> L	20 mm
VoI F >> H	20 mm
TR	3000.0 ms
TE	135.00 ms
Averages	64

Contrast - Common

TR	3000.0 ms
TE	135.00 ms
TM	75.00 ms
Flip Angle	90 deg
Preparation Scans	4
Averages	64
VAPOR	Water Saturation
VAPOR suppr.	Water Suppr.
Water s. BW	135.00 Hz
Water s. Delta Pos.	0.00 ppm

Resolution - Common

Vector Size	2048
Normalize	Off

Geometry - Common

Position	Isocenter
Orientation	Transversal
Rotation	0.00 deg
VoI A >> P	20 mm
VoI R >> L	20 mm
VoI F >> H	20 mm

Geometry - AutoAlign

AutoAlign	---
Initial Position	Isocenter
L	0.0 mm
P	0.0 mm
H	0.0 mm
Initial Orientation	Transversal
Initial Rotation	0.00 deg

System - Miscellaneous

Coil Selection	Auto Coil Select
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Advanced
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off
Adj. Water Suppr.	Off

System - Adjust Volume

Position	Isocenter
Orientation	Transversal
Rotation	0.00 deg
A >> P	20 mm
R >> L	20 mm
F >> H	20 mm
Reset	Off

System - pTx

B1 Shim	TrueForm

System - Tx/Rx

Frequency 1H	123.249442 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	3000.0 ms

Sequence - Common

Sequence Name	steam

Sequence - Common

Preparation Scans	4
Delta Frequency	-1.70 ppm
Measurements	1
Phase Cycling	Auto
Save Uncombined	Off
Bandwidth	2000 Hz
Acquisition Duration	1024 ms
Remove Oversampling	On

Sequence - Nuclei

TX/RX Nucleus	1H
TX/RX Delta Frequency	0 Hz
TX Nucleus	None
TX Delta Frequency	0 Hz

Sequence - Special

RF pulse duration	2560 us
Spoiler max. amplitude	20.00 mT/m
Spoiler amp. ratio	1.50
Refocus grad. factor	1.00
OVS slab thickness	80.00 mm
OVS slab pos. offset	5.00 mm
OVS delta frequency	0.00 ppm
Spoiler duration	1300 us
Acq. window shift	200 us
Min. settling delay	300 us
Gradient ramp time	140 us
VAPOR flip angle	80 deg
VAPOR delay 8	28 ms
VAPOR delay 7	74 ms
OVS pulse duration	5120 us
OVS HS pulse N	1
OVS HS pulse R	40
OVS flip angle RO	90 deg
OVS flip angle PH	90 deg
OVS flip angle SL	90 deg
VAPOR delay 6	68 ms
VAPOR delay 5	106 ms
VAPOR delay 4	105 ms
VAPOR delay 3	146 ms
VAPOR delay 2	100 ms
VAPOR delay 1	150 ms
Enable OVS	On
Resolve averages	Off
Use expt. SAR calc	Off
Check RF clipping	Off
Send ref. scans	Off
Inversion pulse	Off
Symmetric RF pulses	Off
Invert SS grad. pol.	Off
Shift RO frequency	Off
Debug loop type	None

\\Research\Investigators\Frederick\allcustomer_20260909\eja_svs_steam_diff
TA: 3:39 min Coil Selection: Auto Vol: 20×20×20 mm³ Rel. SNR: 1.00

Properties

Start measurement without further preparation	On
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
Graphic segment	Default
Inline Movie	Off

Routine

Position	Isocenter
Orientation	Transversal
Rotation	0.00 deg
Vol A >> P	20 mm
Vol R >> L	20 mm
Vol F >> H	20 mm
TR	3000.0 ms
TE	135.00 ms
Averages	64

Contrast - Common

TR	3000.0 ms
TE	135.00 ms
TM	75.00 ms
Flip Angle	90 deg
Preparation Scans	4
Averages	64
VAPOR	Water Saturation
VAPOR suppr.	Water Suppr.
Water s. BW	135.00 Hz
Water s. Delta Pos.	0.00 ppm

Resolution - Common

Vector Size	2048
Normalize	Off

Geometry - Common

Position	Isocenter
Orientation	Transversal
Rotation	0.00 deg
Vol A >> P	20 mm
Vol R >> L	20 mm
Vol F >> H	20 mm

Geometry - AutoAlign

AutoAlign	---
Initial Position	Isocenter
L	0.0 mm
P	0.0 mm
H	0.0 mm
Initial Orientation	Transversal
Initial Rotation	0.00 deg

System - Miscellaneous

Coil Selection	Auto Coil Select
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Advanced
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off
Adj. Water Suppr.	Off

System - Adjust Volume

Position	Isocenter
Orientation	Transversal
Rotation	0.00 deg
A >> P	20 mm
R >> L	20 mm
F >> H	20 mm
Reset	Off

System - pTx

B1 Shim	TrueForm

System - Tx/Rx

Frequency 1H	123.249442 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	3000.0 ms

Sequence - Common

Sequence Name	steam

Sequence - Common

Preparation Scans	4
Delta Frequency	-1.70 ppm
Measurements	1
Phase Cycling	Auto
Save Uncombined	Off
Bandwidth	2000 Hz
Acquisition Duration	1024 ms
Remove Oversampling	On

Sequence - Nuclei

TX/RX Nucleus	1H
TX/RX Delta Frequency	0 Hz
TX Nucleus	None
TX Delta Frequency	0 Hz

Sequence - Special

RF pulse duration	2560 us
Spoiler max. amplitude	20.00 mT/m
Spoiler amp. ratio	1.50
Refocus grad. factor	1.00
OVS slab thickness	80.00 mm
OVS slab pos. offset	5.00 mm
OVS delta frequency	0.00 ppm
Spoiler duration	1300 us
Acq. window shift	200 us
Min. settling delay	300 us
Gradient ramp time	140 us
VAPOR flip angle	80 deg
VAPOR delay 8	28 ms
VAPOR delay 7	74 ms
OVS pulse duration	5120 us
OVS HS pulse N	1
OVS HS pulse R	40
OVS flip angle RO	90 deg
OVS flip angle PH	90 deg
OVS flip angle SL	90 deg
VAPOR delay 6	68 ms
VAPOR delay 5	106 ms
VAPOR delay 4	105 ms
VAPOR delay 3	146 ms
VAPOR delay 2	100 ms
VAPOR delay 1	150 ms
Diffusion weighting	Off
Enable OVS	On
Resolve averages	Off
Use expt. SAR calc	Off
Check RF clipping	Off
Send ref. scans	Off
Inversion pulse	Off
Symmetric RF pulses	Off
Invert SS grad. pol.	Off
Shift RO frequency	Off
Debug loop type	None

\\Research\Investigators\Frederick\allcustomer_20260909\ep2d_bold_mgh
TA: 8 sec Coil Selection: Auto Voxel Size: 3.9×3.9×5.0 mm³ Acc:: None Rel. SNR: 1.00

Properties

Start measurement without further preparation	On
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
Graphic segment	Default
Inline Movie	Off

Routine

Slice Group	1
Slices	1
Distance Factor	50 %
Position	Isocenter
Orientation	Transversal
Phase Encoding Dir.	A >> P
Phase Oversampling	0 %
FOV Read	500 mm
FOV Phase	100.0 %
Slice Thickness	5.0 mm
TR	223.0 ms
TE	98.00 ms
Averages	1
Concatenations	1
AutoAlign	---

Contrast - Common

TR	223.0 ms
TE	98.00 ms
MTC	Off
Flip Angle	90 deg
Fat-Water Contrast	Fat Saturation
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	20
Delay in TR	0.00 ms

Resolution - Common

FOV Read	500 mm
FOV Phase	100.0 %
Slice Thickness	5.0 mm
Base Resolution	128
Phase Resolution	100 %
Interpolation	Off

Resolution - Acceleration

Acceleration Mode	None
Phase Partial Fourier	Off

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Hamming	Off
Distortion Correction	2D
Static Field Correction	Off
Normalize	Off

Geometry - Common

Slice Group	1
Slices	1
Distance Factor	50 %
Position	Isocenter
Orientation	Transversal
Phase Encoding Dir.	A >> P
Phase Oversampling	0 %
FOV Read	500 mm
FOV Phase	100.0 %
Slice Thickness	5.0 mm
TR	223.0 ms
Multi-Slice Mode	Interleaved
Series	Interleaved
Concatenations	1

Geometry - AutoAlign

Slice Group	1
Position	Isocenter
Orientation	Transversal
Phase Encoding Dir.	A >> P
AutoAlign	---
Initial Position	Isocenter
L	0.0 mm
P	0.0 mm
H	0.0 mm
Initial Orientation	Transversal
Initial Rotation	0.00 deg

Geometry - Saturation

Special Saturation	None

Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Table Position	0 mm

Geometry - Tim Planning Suite

Table Position	H
Inline Composing	Off

System - Miscellaneous

Coil Selection	Auto Coil Select
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Sum of Squares
Matrix Optimization	Off
Coil Focus	Flat

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Standard
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	Isocenter
Orientation	Transversal
Rotation	0.00 deg
A >> P	500 mm
R >> L	500 mm
F >> H	5 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Standard

System - Tx/Rx

Frequency 1H	123.249442 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	223.0 ms
Log Signals	Off
Concatenations	1

BOLD

GLM Statistics	On
Ignore Meas. at Start	0
Ignore After Transition	0
Model Transition States	On
Temp. Highpass Filter	On

BOLD

Threshold	4.00
Paradigm Size	20
Meas[1]	Baseline
Meas[2]	Baseline
Meas[3]	Baseline
Meas[4]	Baseline
Meas[5]	Baseline
Meas[6]	Baseline
Meas[7]	Baseline
Meas[8]	Baseline
Meas[9]	Baseline
Meas[10]	Baseline
Meas[11]	Active
Meas[12]	Active
Meas[13]	Active
Meas[14]	Active
Meas[15]	Active
Meas[16]	Active
Meas[17]	Active
Meas[18]	Active
Meas[19]	Active
Meas[20]	Active
Motion Correction	On
Spatial Filter	Off
Measurements	20
Delay in TR	0.00 ms

Sequence - Part 1

Sequence Name	epfid
Excitation	Standard
RF Pulse Type	Normal
Gradient Mode	Fast
Bandwidth	752 Hz/Px
Echo Spacing	1.38 ms
Free Echo Spacing	Off
EPI Factor	128

Sequence - Part 2

Introduction	Off

Sequence - Special

Imaging Dummy TRs	-1
SMS ACS Dummy TRs	-1
FLEET Dummy Pulses	0
CC Mode	Direct adj coil
RF Clip	0
VERSE Factor	1.00
ACS mode	Standard
Kernel Size	3x3
Wait for TCP/IP Trigger	Off
Reverse Phase Encoding	Off

Sequence - Assistant

SAR Assistant	Off

\\Research\Investigators\Frederick\allcustomer_20260909\ep2d_diff_mgh
TA: 1 sec Coil Selection: Auto Voxel Size: 3.9×3.9×5.0 mm³ Acc:: None Rel. SNR: 1.00

Properties

Start measurement without further preparation	On
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
Graphic segment	Default
Inline Movie	Off

Routine

Slice Group	1
Slices	1
Distance Factor	50 %
Position	Isocenter
Orientation	Transversal
Phase Encoding Dir.	A >> P
Phase Oversampling	0 %
FOV Read	500 mm
FOV Phase	100.0 %
Slice Thickness	5.0 mm
TR	500.0 ms
TE	200.00 ms
Concatenations	1
AutoAlign	---

Contrast - Common

TR	500.0 ms
TE	200.00 ms
MTC	Off
Magn. Preparation	None
Flip Angle 1	90 deg
Flip Angle 2	180 deg
Fat-Water Contrast	Fat Saturation
Fat Saturation	Weak
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Delay in TR	0.00 ms

Resolution - Common

FOV Read	500 mm
FOV Phase	100.0 %
Slice Thickness	5.0 mm

Resolution - Common

Base Resolution	128
Phase Resolution	100 %
Interpolation	Off

Resolution - Acceleration

Acceleration Mode	None
Phase Partial Fourier	6/8

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Distortion Correction	2D
Static Field Correction	Off
Normalize	Off

Geometry - Common

Slice Group	1
Slices	1
Distance Factor	50 %
Position	Isocenter
Orientation	Transversal
Phase Encoding Dir.	A >> P
Phase Oversampling	0 %
FOV Read	500 mm
FOV Phase	100.0 %
Slice Thickness	5.0 mm
TR	500.0 ms
Multi-Slice Mode	Interleaved
Series	Interleaved
Concatenations	1

Geometry - AutoAlign

Slice Group	1
Position	Isocenter
Orientation	Transversal
Phase Encoding Dir.	A >> P
AutoAlign	---
Initial Position	Isocenter
L	0.0 mm
P	0.0 mm
H	0.0 mm
Initial Orientation	Transversal
Initial Rotation	0.00 deg

Geometry - Navigator

Geometry - Saturation

Special Saturation	None

Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Table Position	0 mm
Table Position	H
Inline Composing	Off

System - Miscellaneous

Coil Selection	Auto Coil Select
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Adaptive Combine
Matrix Optimization	Off
Coil Focus	Flat

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Standard
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	Isocenter
Orientation	Transversal
Rotation	0.00 deg
A >> P	500 mm
R >> L	500 mm
F >> H	5 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Standard

System - Tx/Rx

Frequency 1H	123.249442 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	500.0 ms
Concatenations	1

Physio - PACE

Resp. Control	Off
Concatenations	1

Diff

Diffusion Mode	Read
Diff. Directions	1
Diffusion Scheme	Bipolar
Diff. Weightings	1
b-value	0 s/mm²
Averages	1
Dynamic Field Correction	Off
Invert Gray Scale	Off
Diff. Weighted Images	On
Trace Weighted Images	Off
Tensor	Off
FA Maps	Off
ADC Maps	Off
Exponential ADC Maps	Off
Calculated Image	Off

Sequence - Part 1

Sequence Name	epse
Excitation	Standard
RF Pulse Type	Normal
Gradient Mode	Fast
Bandwidth	752 Hz/Px
Echo Spacing	1.38 ms
Free Echo Spacing	Off
Optimization	None
EPI Factor	128

Sequence - Part 2

Introduction	Off
Phase Correction	Internal

Sequence - Special

Imaging Dummy TRs	-1
SMS ACS Dummy TRs	-1
FLEET Dummy Pulses	0
RF Clip	0
VERSE Factor	1.00
ACS mode	Standard
Kernel Size	3x3
Wait for TCP/IP Trigger	Off
Reverse Phase Encoding	Off

Sequence - Assistant

SAR Assistant	Off
Optimization	None

\\Research\Investigators\Frederick\allcustomer_20260909\ep2d_se_sms_mgh
TA: 1 sec Coil Selection: Auto Voxel Size: 3.9×3.9×5.0 mm³ Acc:: None Rel. SNR: 1.00

Properties

Start measurement without further preparation	On
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
Graphic segment	Default
Inline Movie	Off

Routine

Slice Group	1
Slices	1
Distance Factor	50 %
Position	Isocenter
Orientation	Transversal
Phase Encoding Dir.	A >> P
Phase Oversampling	0 %
FOV Read	500 mm
FOV Phase	100.0 %
Slice Thickness	5.0 mm
TR	285.0 ms
TE	182.00 ms
Averages	1
Concatenations	1
AutoAlign	---

Contrast - Common

TR	285.0 ms
TE	182.00 ms
MTC	Off
Magn. Preparation	None
Flip Angle 1	90 deg
Flip Angle 2	180 deg
Fat-Water Contrast	Fat Saturation
Fat Saturation	Weak
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	1
Multiple Series	Off
Delay in TR	0.00 ms

Resolution - Common

FOV Read	500 mm
FOV Phase	100.0 %
Slice Thickness	5.0 mm
Base Resolution	128
Phase Resolution	100 %
Interpolation	Off

Resolution - Acceleration

Acceleration Mode	None
Phase Partial Fourier	Off

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Hamming	Off
Distortion Correction	2D
Static Field Correction	Off
Normalize	Off

Geometry - Common

Slice Group	1
Slices	1
Distance Factor	50 %
Position	Isocenter
Orientation	Transversal
Phase Encoding Dir.	A >> P
Phase Oversampling	0 %
FOV Read	500 mm
FOV Phase	100.0 %
Slice Thickness	5.0 mm
TR	285.0 ms
Multi-Slice Mode	Interleaved
Series	Interleaved
Concatenations	1

Geometry - AutoAlign

Slice Group	1
Position	Isocenter
Orientation	Transversal
Phase Encoding Dir.	A >> P
AutoAlign	---
Initial Position	Isocenter
L	0.0 mm
P	0.0 mm
H	0.0 mm
Initial Orientation	Transversal
Initial Rotation	0.00 deg

Geometry - Saturation

Special Saturation	None

Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Table Position	0 mm
Table Position	H
Inline Composing	Off

System - Miscellaneous

Coil Selection	Auto Coil Select
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Sum of Squares
Matrix Optimization	Off
Coil Focus	Flat

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Standard
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	Isocenter
Orientation	Transversal
Rotation	0.00 deg
A >> P	500 mm
R >> L	500 mm
F >> H	5 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Standard

System - Tx/Rx

Frequency 1H	123.249442 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	285.0 ms
Concatenations	1

Inline - Subtraction

Subtract	Off
Measurements	1
StdDev	Off
Save Original Images	On

Inline - MIP

MIP Sag	Off
MIP Cor	Off
MIP Tra	Off
MIP Time	Off
Radial MIP	Off
Save Original Images	On
MPR Sag	Off
MPR Cor	Off
MPR Tra	Off

Inline - Composing

Inline Composing	Off

Sequence - Part 1

Sequence Name	epse
Excitation	Standard
RF Pulse Type	Normal
Gradient Mode	Fast
Bandwidth	752 Hz/Px
Echo Spacing	1.38 ms
Free Echo Spacing	Off
EPI Factor	128

Sequence - Part 2

Introduction	Off

Sequence - Special

Imaging Dummy TRs	-1
SMS ACS Dummy TRs	-1
FLEET Dummy Pulses	0
RF Clip	0
VERSE Factor	1.00
ACS mode	Standard
Kernel Size	3x3
Wait for TCP/IP Trigger	Off
Reverse Phase Encoding	Off

Sequence - Assistant

SAR Assistant	Off

\\Research\Investigators\Frederick\allcustomer_20260909\ep_moco_nav_set_ABCD
TA: 1 sec Coil Selection: Auto Voxel Size: 3.9×3.9×5.0 mm³ Rel. SNR: 1.00

Properties

Start measurement without further preparation	On
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
Graphic segment	Default
Inline Movie	Off

Routine

Slice Group	1
Slices	1
Distance Factor	50 %
Position	Isocenter
Orientation	Transversal
Phase Encoding Dir.	A >> P
Phase Oversampling	0 %
FOV Read	500 mm
FOV Phase	100.0 %
Slice Thickness	5.0 mm
TR	44.0 ms
TE	15.00 ms
Averages	1
Concatenations	1
AutoAlign	---

Contrast - Common

TR	44.0 ms
TE	15.00 ms
MTC	Off
Flip Angle	90 deg
Fat-Water Contrast	Fat Saturation
Contrasts	1
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	1
Multiple Series	Each Measurement

Resolution - Common

FOV Read	500 mm
FOV Phase	100.0 %
Slice Thickness	5.0 mm

Resolution - Common

Base Resolution	128
Phase Resolution	100 %
Interpolation	Off

Resolution - Acceleration

Phase Partial Fourier	Off

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Hamming	Off
Distortion Correction	Off
Normalize	Off

Geometry - Common

Slice Group	1
Slices	1
Distance Factor	50 %
Position	Isocenter
Orientation	Transversal
Phase Encoding Dir.	A >> P
Phase Oversampling	0 %
FOV Read	500 mm
FOV Phase	100.0 %
Slice Thickness	5.0 mm
TR	44.0 ms
Multi-Slice Mode	Interleaved
Series	Interleaved
Concatenations	1

Geometry - AutoAlign

Slice Group	1
Position	Isocenter
Orientation	Transversal
Phase Encoding Dir.	A >> P
AutoAlign	---
Initial Position	Isocenter
L	0.0 mm
P	0.0 mm
H	0.0 mm
Initial Orientation	Transversal
Initial Rotation	0.00 deg

Geometry - Saturation

Special Saturation	None

Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Table Position	0 mm
Table Position	H

Geometry - Tim Planning Suite

Inline Composing	Off

System - Miscellaneous

Coil Selection	Auto Coil Select
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Sum of Squares
Matrix Optimization	Off
Coil Focus	Flat

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Standard
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	Isocenter
Orientation	Transversal
Rotation	0.00 deg
A >> P	500 mm
R >> L	500 mm
F >> H	5 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Slice-sel.

System - Tx/Rx

Frequency 1H	123.249442 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	44.0 ms
Concatenations	1

Sequence - Part 1

Sequence Name	ABCD
Dimension	2D
Excitation	Slice-sel.
RF Pulse Type	Normal
Gradient Mode	Fast
Flow Compensation	None
Bandwidth	752 Hz/Px

Sequence - Part 1

Echo Spacing	1.44 ms
Free Echo Spacing	Off
EPI Factor	16

Sequence - Part 2

Introduction	Off
RF Spoiling	On

Sequence - Special

Protocol filename	Generic

Sequence - Assistant

SAR Assistant	Off

\\Research\Investigators\Frederick\allcustomer_20260909\fastestmap
TA: 12 sec Coil Selection: Auto Vol: 20×20×20 mm³ Rel. SNR: 1.00

Properties

Start measurement without further preparation	On
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
Graphic segment	Default
Inline Movie	Off

Routine

Position	Isocenter
Orientation	Transversal
Rotation	0.00 deg
Vol A >> P	20 mm
Vol R >> L	20 mm
Vol F >> H	20 mm
TR	2000.0 ms
TE	49.20 ms
Averages	1

Contrast - Common

TR	2000.0 ms
TE	49.20 ms
Tau	5.00 ms
Excite flip angle	90 deg
Refocus flip angle	180 deg
Averages	1
Water Suppression	None

Resolution - Common

Vector Size	256
Normalize	Off

Geometry - Common

Position	Isocenter
Orientation	Transversal
Rotation	0.00 deg
Vol A >> P	20 mm
Vol R >> L	20 mm
Vol F >> H	20 mm

Geometry - AutoAlign

AutoAlign	---
Initial Position	Isocenter

Geometry - AutoAlign

L	0.0 mm
P	0.0 mm
H	0.0 mm
Initial Orientation	Transversal
Initial Rotation	0.00 deg

System - Miscellaneous

Coil Selection	Auto Coil Select
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Advanced
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off
Adj. Water Suppr.	Off

System - Adjust Volume

Position	Isocenter
Orientation	Transversal
Rotation	0.00 deg
A >> P	20 mm
R >> L	20 mm
F >> H	20 mm
Reset	Off

System - pTx

B1 Shim	TrueForm

System - Tx/Rx

Frequency 1H	123.249442 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	2000.0 ms

Sequence - Common

Sequence Name	fastmp
Delta Frequency	0.00 ppm
Measurements	1

Sequence - Common

Phase Cycling	None
Save Uncombined	Off
Bandwidth	100000 Hz
Acquisition Duration	2 ms

Sequence - Special

Type of fit	Full 6-proj
VoI fit factor	100.00 %
Force spherical fit VoI	Off Force spherical fit VoI
Save plots to database	Off
Refocus pulses	Normal
Excite pulse duration	6400 us
Refocus pulse duration	6400 us
Bar FoV	384 mm
Bar thickness	10.00 mm
Inversion pulse	Off
Multi-echo acquisition	On
Number of echoes	4

\\Research\Investigators\Frederick\allcustomer_20260909\NoiseSensitivityMap
TA: 5 sec Coil Selection: Auto Voxel Size: 1.2×1.2×5.0 mm³ Rel. SNR: 1.00

Properties

Start measurement without further preparation	On
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
Graphic segment	Default
Inline Movie	Off

Routine

Slice Group	1
Slices	1
Distance Factor	20 %
Position	Isocenter
Orientation	Transversal
Phase Encoding Dir.	A >> P
Phase Oversampling	0 %
FOV Read	300 mm
FOV Phase	100.0 %
Slice Thickness	5.0 mm
TR	9.1 ms
TE	4.80 ms
Averages	1
Concatenations	1
AutoAlign	---

Contrast - Common

TR	9.1 ms
TE	4.80 ms
MTC	Off
Flip Angle	15 deg
Fat-Water Contrast	Standard
Contrasts	1
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	1
Multiple Series	Off

Resolution - Common

FOV Read	300 mm
FOV Phase	100.0 %
Slice Thickness	5.0 mm

Resolution - Common

Base Resolution	256
Phase Resolution	100 %
Interpolation	Off

Resolution - Acceleration

Phase Partial Fourier	Off

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Distortion Correction	2D
Normalize	Off
Image Filter	Off

Geometry - Common

Slice Group	1
Slices	1
Distance Factor	20 %
Position	Isocenter
Orientation	Transversal
Phase Encoding Dir.	A >> P
Phase Oversampling	0 %
FOV Read	300 mm
FOV Phase	100.0 %
Slice Thickness	5.0 mm
TR	9.1 ms
Multi-Slice Mode	Sequential
Series	Ascending
Concatenations	1

Geometry - AutoAlign

Slice Group	1
Position	Isocenter
Orientation	Transversal
Phase Encoding Dir.	A >> P
AutoAlign	---
Initial Position	Isocenter
L	0.0 mm
P	0.0 mm
H	0.0 mm
Initial Orientation	Transversal
Initial Rotation	0.00 deg

Geometry - Saturation

Special Saturation	None

Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Table Position	0 mm
Table Position	H

Geometry - Tim Planning Suite

Inline Composing	Off

System - Miscellaneous

Coil Selection	Auto Coil Select
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Adaptive Combine
Matrix Optimization	Off
Coil Focus	Flat

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Tune up
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	Isocenter
Orientation	Transversal
Rotation	0.00 deg
A >> P	263 mm
R >> L	350 mm
F >> H	350 mm
Reset	Off

System - pTx

B1 Shim	TrueForm

System - Tx/Rx

Frequency 1H	123.249442 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	9.1 ms
Concatenations	1

Inline - Subtraction

Subtract	Off
Measurements	1
StdDev	Off
Save Original Images	On

Inline - MIP

MIP Sag	Off
MIP Cor	Off
MIP Tra	Off
MIP Time	Off
Radial MIP	Off
Save Original Images	On
MPR Sag	Off
MPR Cor	Off
MPR Tra	Off

Inline - Composing

Inline Composing	Off

Sequence - Part 1

Sequence Name	fl
Dimension	2D
Gradient Mode	Fast
Bandwidth	390 Hz/Px

Sequence - Part 2

Introduction	Off
RF Spoiling	On

Sequence - Nuclei

TX/RX Nucleus	1H
TX/RX Delta Frequency	0 Hz
TX Nucleus	None
TX Delta Frequency	0 Hz

Sequence - Special

ICE program	CoilArrayUtil
number of noise lines	384 lines
Optimal SNR	On
GFactor	On
Condition number	Off
Rx coil diode switching	On
coil channel reordering	Off

Sequence - Assistant

SAR Assistant	Off

\\Research\Investigators\Frederick\allcustomer_20260909\rslh_ep3d_vaso
TA: 14 sec Coil Selection: Auto Voxel Size: 2.5×2.5×2.5 mm³ Acc:: None Rel. SNR: 1.00

Properties

Start measurement without further preparation	On
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
Graphic segment	Default
Inline Movie	Off

Routine

Slab Group	1
Slabs	1
Position	Isocenter
Orientation	Transversal
Phase Encoding Dir.	A >> P
Slices per Slab	48
Phase Oversampling	0.0 %
Slice Oversampling	0.0 %
FOV Read	210 mm
FOV Phase	100.0 %
Slice Thickness	2.50 mm
TR	130.0 ms
Vol. TR	12.50064 s
TE 1	48.00 ms
Averages	1
Multi-echo Shots	1
AutoAlign	---

Contrast - Common

TR	130.0 ms
Vol. TR	12.50064 s
TE 1	48.00 ms
Multi-echo spacing	88.00 ms
MTC	Off
Magn. Preparation	Non-sel. IR
TI 1	3130.32 ms
TI 2	9370.32 ms
Flip Angle	13 deg
Fat-Water Contrast	Standard
Contrasts	1
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	1

Contrast - Dynamic

Reordering	Linear

Resolution - Common

FOV Read	210 mm
FOV Phase	100.0 %
Slice Thickness	2.50 mm
Base Resolution	84
Phase Resolution	100 %
Slice Resolution	100 %
Interpolation	Off

Resolution - Acceleration

Acceleration Mode	None
Phase Partial Fourier	Off
Slice Partial Fourier	Off
Asymmetric Echo	Off

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Distortion Correction	Off
Normalize	Off
Image Filter	Off

Geometry - Common

Slab Group	1
Slabs	1
Position	Isocenter
Orientation	Transversal
Phase Encoding Dir.	A >> P
Slices per Slab	48
Phase Oversampling	0.0 %
Slice Oversampling	0.0 %
FOV Read	210 mm
FOV Phase	100.0 %
Slice Thickness	2.50 mm
TR	130.0 ms
Vol. TR	12.50064 s
Multi-echo Shots	1

Geometry - AutoAlign

Slab Group	1
Position	Isocenter
Orientation	Transversal
Phase Encoding Dir.	A >> P
AutoAlign	---
Initial Position	Isocenter
L	0.0 mm
P	0.0 mm
H	0.0 mm

Geometry - AutoAlign

Initial Orientation	Transversal
Initial Rotation	0.00 deg

Geometry - Saturation

Saturation Mode	Standard

Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Table Position	0 mm
Table Position	H
Inline Composing	Off

System - Miscellaneous

Coil Selection	Auto Coil Select
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Sum of Squares
Matrix Optimization	Off
Coil Focus	Flat

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Brain
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	Isocenter
Orientation	Transversal
Rotation	0.00 deg
A >> P	210 mm
R >> L	210 mm
F >> H	120 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Slab-sel.

System - Tx/Rx

Frequency 1H	123.249442 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Sequence - Part 1

Sequence Name	vso rslh6.0
Dimension	3D

Sequence - Part 1

Excitation	Slab-sel.
RF Pulse Type	Normal
Gradient Mode	Fast
Reordering	Linear
Bandwidth	1044 Hz/Px
Echo Spacing	1.04 ms
Asymmetric Echo	Off
Turbo Factor	48
Segmentation	1
EPI Factor	84

Sequence - Part 2

Introduction	On
RF Spoiling	On

Sequence - Special

RF duration	1080 us
RF time x BW	18
PAT ref. FA	5 deg
Fat sat. FA	110 deg
Variable FA	4
Invert PE	Off
Min. TE w/ PF	On
Expert ramping	Off
Trigger per shot	Off
Noise image	Off
Relax spoilers	Off
MT flip angle	170 deg
MT off-res.	2000 Hz
MT RF duration	12800 us
Phase Correction	per Series
EPI ramp factor	1.10
EPI ramp factor 2	1.10
Ramp sampling	1.00 1:max
Slab Scale	0 %
Modify Ice Config	On
G-factor map	Off
Mosaic DICOMs	Off
GRAPPA Regularization	5000 /10^6
G. spoil dephasing[1]	0 pi
G. spoil dephasing[2]	2 pi
G. spoil dephasing[3]	2 pi
Read polarity	Regular RO
VASO Pre-TI1 delay	0.00 ms
VASO Pre-TI2 delay	0 ms
VASO Pre-Inv delay	0 ms

Sequence - Assistant

SAR Assistant	Off

\\Research\Investigators\Frederick\allcustomer_20260909\space_mgh_epinav_ABCD
TA: 6:47 min Coil Selection: Auto Voxel Size: 1.2×1.2×3.0 mm³ Acc:: None Rel. SNR: 1.00

Properties

Start measurement without further preparation	On
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
Graphic segment	Default
Inline Movie	Off

Routine

Slab Group	1
Slabs	1
Position	Isocenter
Orientation	Transversal
Phase Encoding Dir.	A >> P
Slices per Slab	64
Phase Oversampling	0 %
Slice Oversampling	0.0 %
FOV Read	300 mm
FOV Phase	100.0 %
Slice Thickness	3.00 mm
TR	3000.0 ms
TE	122.00 ms
Averages	1.0
Concatenations	1
AutoAlign	---

Contrast - Common

TR	3000.0 ms
TE	122.00 ms
MTC	Off
Magn. Preparation	None
Flip Angle Mode	T2 Var
Fat-Water Contrast	Standard
Dark Blood	Off
Blood Suppression	Off
Wrap-up Magn.	None
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	1
Multiple Series	Each Measurement
Reordering	Linear

Resolution - Common

FOV Read	300 mm
FOV Phase	100.0 %
Slice Thickness	3.00 mm
Base Resolution	256
Phase Resolution	100 %
Slice Resolution	100 %
Interpolation	Off

Resolution - Acceleration

Acceleration Mode	None
Phase Partial Fourier	Allowed
Slice Partial Fourier	Off
Elliptical Scanning	On

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Distortion Correction	Off
Normalize	Off
Image Filter	Off

Geometry - Common

Slab Group	1
Slabs	1
Position	Isocenter
Orientation	Transversal
Phase Encoding Dir.	A >> P
Slices per Slab	64
Phase Oversampling	0 %
Slice Oversampling	0.0 %
FOV Read	300 mm
FOV Phase	100.0 %
Slice Thickness	3.00 mm
TR	3000.0 ms
Concatenations	1

Geometry - AutoAlign

Slab Group	1
Position	Isocenter
Orientation	Transversal
Phase Encoding Dir.	A >> P
AutoAlign	---
Initial Position	Isocenter
L	0.0 mm
P	0.0 mm
H	0.0 mm
Initial Orientation	Transversal
Initial Rotation	0.00 deg

Geometry - Navigator

Geometry - Saturation

Special Saturation	None

Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Table Position	0 mm
Table Position	H
Inline Composing	Off

System - Miscellaneous

Coil Selection	Auto Coil Select
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Adaptive Combine
Matrix Optimization	Off
Coil Focus	Flat

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Tune up
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	Isocenter
Orientation	Transversal
Rotation	0.00 deg
A >> P	263 mm
R >> L	350 mm
F >> H	350 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Non-sel.

System - Tx/Rx

Frequency 1H	123.249442 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000
Gain	High

Physio - Signal

1st Signal/Mode	None
Trigger Delay	0 ms
TR	3000.0 ms
Concatenations	1

Physio - Cardiac

Fat-Water Contrast	Standard
Magn. Preparation	None
Dark Blood	Off
FOV Read	300 mm
FOV Phase	100.0 %
Phase Resolution	100 %

Physio - PACE

Resp. Control	Off
Concatenations	1

Inline - Subtraction

Subtract	Off
Measurements	1
StdDev	Off
Save Original Images	On

Inline - MIP

MIP Sag	Off
MIP Cor	Off
MIP Tra	Off
MIP Time	Off
Radial MIP	Off
Save Original Images	On
MPR Sag	Off
MPR Cor	Off
MPR Tra	Off

Inline - Composing

Inline Composing	Off

Sequence - Part 1

Sequence Name	spc
Dimension	3D
Excitation	Non-sel.
RF Pulse Type	Normal
Gradient Mode	Fast
Flow Compensation	None
Reordering	Linear
Bandwidth	501 Hz/Px
Echo Spacing	3.92 ms
Turbo Factor	96
Echo Train Duration	306 ms

Sequence - Part 2

Introduction	On

Sequence - Special

Include Nav.	Off
Apply moco to	parent and nav
Remeasure	0 TRs
Reacq. threshold	0.50
Feedback Delay	80 ms
Moco ref. image	Use Temp Ref

Sequence - Special

K-space streaming	None
ABCD navigator	Off
Apply freq to	parent and nav

Sequence - Assistant

SAR Assistant	Off
Allowed Delay	0 s

\\Research\Investigators\Frederick\allcustomer_20260909\tfl_mgh_epinav_ABCD
TA: 18 sec Coil Selection: Auto Voxel Size: 4.7×4.7×8.0 mm³ Acc:: None Rel. SNR: 1.00

Properties

Start measurement without further preparation	On
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
Graphic segment	Default
Inline Movie	Off

Routine

Slice Group	1
Slices	1
Distance Factor	50 %
Position	Isocenter
Orientation	Transversal
Phase Encoding Dir.	A >> P
Phase Oversampling	0 %
FOV Read	300 mm
FOV Phase	100.0 %
Slice Thickness	8.0 mm
TR	4000.0 ms
TE	1.39 ms
Averages	1
Concatenations	1
AutoAlign	---

Contrast - Common

TR	4000.0 ms
TE	1.39 ms
Magn. Preparation	Slice-sel. IR
TI	1000 ms
Flip Angle	8 deg
Fat-Water Contrast	Standard
Dark Blood	Off
Contrasts	1
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	1
Multiple Series	Each Measurement

Resolution - Common

FOV Read	300 mm

Resolution - Common

FOV Phase	100.0 %
Slice Thickness	8.0 mm
Base Resolution	64
Phase Resolution	100 %
Interpolation	Off

Resolution - Acceleration

Acceleration Mode	None
Phase Partial Fourier	Off
Asymmetric Echo	Off

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Distortion Correction	Off
Normalize	Off
Image Filter	Off

Geometry - Common

Slice Group	1
Slices	1
Distance Factor	50 %
Position	Isocenter
Orientation	Transversal
Phase Encoding Dir.	A >> P
Phase Oversampling	0 %
FOV Read	300 mm
FOV Phase	100.0 %
Slice Thickness	8.0 mm
TR	4000.0 ms
Multi-Slice Mode	Single Shot
Series	Interleaved
Concatenations	1

Geometry - AutoAlign

Slice Group	1
Position	Isocenter
Orientation	Transversal
Phase Encoding Dir.	A >> P
AutoAlign	---
Initial Position	Isocenter
L	0.0 mm
P	0.0 mm
H	0.0 mm
Initial Orientation	Transversal
Initial Rotation	0.00 deg

Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Table Position	0 mm

Geometry - Tim Planning Suite

Table Position	H
Inline Composing	Off

System - Miscellaneous

Coil Selection	Auto Coil Select
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Adaptive Combine
Matrix Optimization	Off
Coil Focus	Flat

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Tune up
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	Isocenter
Orientation	Transversal
Rotation	0.00 deg
A >> P	263 mm
R >> L	350 mm
F >> H	350 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Slice-sel.

System - Tx/Rx

Frequency 1H	123.249442 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	4000.0 ms
Concatenations	1

Physio - Cardiac

Fat-Water Contrast	Standard
Magn. Preparation	Slice-sel. IR
TI	1000 ms
Dark Blood	Off
FOV Read	300 mm
FOV Phase	100.0 %

Physio - Cardiac

Phase Resolution	100 %

Inline - Subtraction

Subtract	Off
Measurements	1
StdDev	Off
Save Original Images	On

Inline - MIP

MIP Sag	Off
MIP Cor	Off
MIP Tra	Off
MIP Time	Off
Radial MIP	Off
Save Original Images	On
MPR Sag	Off
MPR Cor	Off
MPR Tra	Off

Inline - Composing

Inline Composing	Off

Sequence - Part 1

Sequence Name	tfl_vn
Dimension	2D
Excitation	Slice-sel.
RF Pulse Type	Fast
Gradient Mode	Fast
Flow Compensation	None
Bandwidth	650 Hz/Px
Echo Spacing	2.78 ms
Asymmetric Echo	Off
Turbo Factor	64

Sequence - Part 2

Introduction	On
RF Spoiling	On
Incr. Gradient Spoiling	Off

Sequence - Special

Readout polarity	Positive
Nav. location	None
Apply moco to	parent and nav
Remeasure	0 TRs
Reacq. threshold	0.50
Feedback Delay	80 ms
Moco ref. image	Use Temp Ref
K-space streaming	None
ABCD navigator	Off
Add. grad time	0.00 ms
Apply freq to	parent and nav
Averaging	None

Sequence - Assistant

SAR Assistant	Off

\\Research\Investigators\Frederick\allcustomer_20260909\tfl_mgh_multiecho
TA: 4 sec Coil Selection: Auto Voxel Size: 4.7×4.7×8.0 mm³ Acc:: None Rel. SNR: 1.00

Properties

Start measurement without further preparation	On
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
Graphic segment	Default
Inline Movie	Off

Routine

Slice Group	1
Slices	1
Distance Factor	50 %
Position	Isocenter
Orientation	Transversal
Phase Encoding Dir.	A >> P
Phase Oversampling	0 %
FOV Read	300 mm
FOV Phase	100.0 %
Slice Thickness	8.0 mm
TR	2000.0 ms
TE	1.39 ms
Averages	1
Concatenations	1
AutoAlign	---

Contrast - Common

TR	2000.0 ms
TE	1.39 ms
Magn. Preparation	Slice-sel. IR
TI	300 ms
Flip Angle	8 deg
Fat-Water Contrast	Standard
Dark Blood	Off
Contrasts	1
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	1
Multiple Series	Each Measurement

Resolution - Common

FOV Read	300 mm

Resolution - Common

FOV Phase	100.0 %
Slice Thickness	8.0 mm
Base Resolution	64
Phase Resolution	100 %
Interpolation	Off

Resolution - Acceleration

Acceleration Mode	None
Phase Partial Fourier	Off
Asymmetric Echo	Off

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Distortion Correction	Off
Normalize	Off
Image Filter	Off

Geometry - Common

Slice Group	1
Slices	1
Distance Factor	50 %
Position	Isocenter
Orientation	Transversal
Phase Encoding Dir.	A >> P
Phase Oversampling	0 %
FOV Read	300 mm
FOV Phase	100.0 %
Slice Thickness	8.0 mm
TR	2000.0 ms
Multi-Slice Mode	Single Shot
Series	Interleaved
Concatenations	1

Geometry - AutoAlign

Slice Group	1
Position	Isocenter
Orientation	Transversal
Phase Encoding Dir.	A >> P
AutoAlign	---
Initial Position	Isocenter
L	0.0 mm
P	0.0 mm
H	0.0 mm
Initial Orientation	Transversal
Initial Rotation	0.00 deg

Geometry - Navigator

Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Table Position	0 mm
Table Position	H
Inline Composing	Off

System - Miscellaneous

Coil Selection	Auto Coil Select
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Adaptive Combine
Matrix Optimization	Off
Coil Focus	Flat

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Tune up
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	Isocenter
Orientation	Transversal
Rotation	0.00 deg
A >> P	263 mm
R >> L	350 mm
F >> H	350 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Slice-sel.

System - Tx/Rx

Frequency 1H	123.249442 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	2000.0 ms
Concatenations	1

Physio - Cardiac

Fat-Water Contrast	Standard
Magn. Preparation	Slice-sel. IR
TI	300 ms
Dark Blood	Off

Physio - Cardiac

FOV Read	300 mm
FOV Phase	100.0 %
Phase Resolution	100 %

Physio - PACE

Resp. Control	Off
Concatenations	1

Inline - Subtraction

Subtract	Off
Measurements	1
StdDev	Off
Save Original Images	On

Inline - MIP

MIP Sag	Off
MIP Cor	Off
MIP Tra	Off
MIP Time	Off
Radial MIP	Off
Save Original Images	On
MPR Sag	Off
MPR Cor	Off
MPR Tra	Off

Inline - Composing

Inline Composing	Off

Sequence - Part 1

Sequence Name	tfl_me
Dimension	2D
Excitation	Slice-sel.
RF Pulse Type	Fast
Gradient Mode	Fast
Flow Compensation	None
Bandwidth	650 Hz/Px
Echo Spacing	2.78 ms
Asymmetric Echo	Off
Turbo Factor	64

Sequence - Part 2

Introduction	On
RF Spoiling	On
Incr. Gradient Spoiling	Off

Sequence - Special

Readout polarity	Positive
Readout trajectory	Bipolar
Gradient spoiling	Siemens
Gradient moment factor	1.00
Averaging	None

Sequence - Assistant

SAR Assistant	Off

\\Research\Investigators\Frederick\allcustomer_20260909\ZPL_RG_EPSI_FID_v1h
TA: 4:33:55 h Coil Selection: Auto Voxel Size: 1.1×2.2×1.0 mm³ Rel. SNR: 1.00

Properties

Start measurement without further preparation	On
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
Graphic segment	Default
Inline Movie	Off

Routine

Position	Isocenter
Orientation	Transversal
Rotation	0.00 deg
FOV A >> P	240 mm
FOV R >> L	240 mm
FOV F >> H	72 mm
Vol A >> P	240 mm
Vol R >> L	240 mm
Vol F >> H	60 mm
TR	1000.0 ms
TE	1.60 ms
Averages	1

Contrast - Common

TR	1000.0 ms
TE	1.60 ms
Ernst angle	61 deg
Prep. pulses	6 #
T1 Estimated	1400 ms
T1 WaterSupp	1000 ms
WaterSupp Angle	90 deg
FOV Pos, Read	0 um
FOV Pos, Phase	0 um
FOV Pos, Slice	0 um
Flip Angle	27 deg
Preparation Scans	10
Averages	1
Water Suppression	
Water Suppr. BW	80 Hz

Resolution - Common

FOV A >> P	240 mm
FOV R >> L	240 mm
FOV F >> H	72 mm
Scan Res. A >> P	216

Resolution - Common

Scan Res. R >> L	110
Scan Res. F >> H	72
Interpol. Res. A >> P	256
Interpol. Res. R >> L	128
Interpol. Res. F >> H	128
Hamming	Off
Dimension	3D
Phase Encoding	Full
Normalize	Off

Geometry - Common

Position	Isocenter
Orientation	Transversal
Rotation	0.00 deg
FOV A >> P	240 mm
FOV R >> L	240 mm
FOV F >> H	72 mm
Vol A >> P	240 mm
Vol R >> L	240 mm
Vol F >> H	60 mm
Fully Excited Vol	Off
Saturation Region	1
Thickness	30.00 mm
Position	Isocenter
Orientation	Transversal
Sat. Delta Frequ.	0.00 ppm
Saturation Region	2
Thickness	30.00 mm
Position	Isocenter
Orientation	Transversal
Sat. Delta Frequ.	0.00 ppm

Geometry - AutoAlign

Slab Group	1
Position	Isocenter
Orientation	Transversal
Phase Encoding Dir.	A >> P
AutoAlign	---
Initial Position	Isocenter
L	0.0 mm
P	0.0 mm
H	0.0 mm
Initial Orientation	Transversal
Initial Rotation	0.00 deg

System - Miscellaneous

Coil Selection	Auto Coil Select
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L

System - Miscellaneous

Coronal	A >> P
Transversal	F >> H

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Advanced
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off
Adj. Water Suppr.	Off

System - Adjust Volume

Position	Isocenter
Orientation	Transversal
Rotation	0.00 deg
A >> P	240 mm
R >> L	240 mm
F >> H	60 mm
Reset	Off

System - pTx

B1 Shim	TrueForm

System - Tx/Rx

Frequency 1H	123.249442 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Sequence - Common

Sequence Name	ZPL_epsi_fid_v1h
Preparation Scans	10
Delta Frequency	-2.40 ppm
Dimension	3D
Save Uncombined	Off
Bandwidth	167000 Hz
Acquisition Duration	0 ms
Remove Oversampling	Off

Sequence - Nuclei

TX/RX Nucleus	1H
TX/RX Delta Frequency	0 Hz
TX Nucleus	None
TX Delta Frequency	0 Hz

Sequence - Special

PE Samp EPSI	No Customized Sampling Pattern
PE Samp EBSI	No Customized Sampling Pattern
PE Samp DEPI	No Customized Sampling Pattern

Sequence - Special

Mode: DEPInav	0 #
Mode: DEPIref	0 #
Mode: MotionNav	60 #
Mode: WatSupNav	1 #
Mode: BipolNav	1 #
Mode: RFspoil	0 #
Ramping time (PE)	120 us
Grad duration (PE)	600 us
Blip ramp time (PE)	60 us
Blip ramp time (SL)	60 us
Jump ramp time (JP)	170 us
Jump flat time (JP)	200 us
Ramping time (SP)	180 us
Spoil duration (SP)	4000 us
Spoil amplitude(SP)	15 mT/mm
EPSI. Num: Echo	74 #
EPSI. Num: Shift	0 #
EPSI. Ramp. Samp	16 #
EPSI. RO. Ramp time	110 us
EPSI. EchoSpace	1760 us
EPSC. Num: Read	110 #
EPSC. Num: Phas	24 #
EPSC. Num: Slic	24 #
EPSC. Num: Echo	48 #
EPSC. Ramp. Samp	16 #
EPSC2. Num: Phas	6 #
EPSC2. Num: Slic	6 #
EPSC. EchoSpace	1760 us
Fnav. Read points	100 #
Fnav. Dwell time	10 us
Mnav. Echo Pair	10 #

\\Research\Investigators\Frederick\allcustomer_20260909\ZPL_RG_EPSI_FID_v2e
TA: 3:54:22 h Coil Selection: Auto Voxel Size: 1.1×1.0×1.0 mm³ Rel. SNR: 1.00

Properties

Start measurement without further preparation	On
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
Graphic segment	Default
Inline Movie	Off

Routine

Position	Isocenter
Orientation	Transversal
Rotation	0.00 deg
FOV A >> P	240 mm
FOV R >> L	240 mm
FOV F >> H	72 mm
Vol A >> P	240 mm
Vol R >> L	240 mm
Vol F >> H	60 mm
TR	1000.0 ms
TE	1.60 ms
Averages	1

Contrast - Common

TR	1000.0 ms
TE	1.60 ms
Ernst angle	61 deg
Prep. pulses	6 #
T1 Estimated	1400 ms
Flip Angle	32 deg
Preparation Scans	0
Averages	1
Water Suppression	None

Resolution - Common

FOV A >> P	240 mm
FOV R >> L	240 mm
FOV F >> H	72 mm
Scan Res. A >> P	216
Scan Res. R >> L	230
Scan Res. F >> H	72
Interpol. Res. A >> P	256
Interpol. Res. R >> L	256
Interpol. Res. F >> H	128
Hamming	Off

Resolution - Common

Dimension	3D
Phase Encoding	Full
Normalize	Off

Geometry - Common

Position	Isocenter
Orientation	Transversal
Rotation	0.00 deg
FOV A >> P	240 mm
FOV R >> L	240 mm
FOV F >> H	72 mm
Vol A >> P	240 mm
Vol R >> L	240 mm
Vol F >> H	60 mm
Fully Excited Vol	Off

Geometry - AutoAlign

Slab Group	1
Position	Isocenter
Orientation	Transversal
Phase Encoding Dir.	A >> P
AutoAlign	---
Initial Position	Isocenter
L	0.0 mm
P	0.0 mm
H	0.0 mm
Initial Orientation	Transversal
Initial Rotation	0.00 deg

System - Miscellaneous

Coil Selection	Auto Coil Select
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Advanced
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off
Adj. Water Suppr.	Off

System - Adjust Volume

Position	Isocenter
Orientation	Transversal

System - Adjust Volume

Rotation	0.00 deg
A >> P	240 mm
R >> L	240 mm
F >> H	60 mm
Reset	Off

System - pTx

B1 Shim	TrueForm

System - Tx/Rx

Frequency 1H	123.249442 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Sequence - Common

Sequence Name	ZPL_epsi_fid
Preparation Scans	0
Delta Frequency	0.00 ppm
Dimension	3D
Save Uncombined	Off
Acquisition Delay	2.30 ms
Bandwidth	167000 Hz
Acquisition Duration	0 ms
Remove Oversampling	Off

Sequence - Nuclei

TX/RX Nucleus	1H
TX/RX Delta Frequency	0 Hz
TX Nucleus	None
TX Delta Frequency	0 Hz

Sequence - Special

PE Samp EPSI	No Customized Sampling Pattern
PE Samp Prep	No Customized Sampling Pattern
Mode: T2-Prep	9 #
Mode: MotionNav	0 #
Mode: WatSupNav	0 #
Mode: BipolNav	1 #
Mode: RFspoil	0 #
Ramping time (PE)	120 us
Grad duration (PE)	620 us
Blip ramp time (PE)	60 us
Blip ramp time (SL)	60 us
Spoil duration (SP)	4000 us
EPSI. Num: Echo	18 #
EPSI. Echo: Shift	0 #
EPSI. BlipLeng: Phas	256 #
EPSI. BlipStep: Phas	3 #
EPSI. BlipStep: Slic	1 #
EPSI. Ramp. Samp	16 #
EPSI. RO. Ramp time	110 us

Sequence - Special

EPSI. EchoSpace	3200 us
T2Prep. Echo time	100000 us
T2Prep. Pulse mode	1 #
Prep. Num Wait TR	4 #
Pulse duration (RF)	1200 us
Pulse bandwidth (RF)	7291 Hz
FOV Pos, Read	0 um
FOV Pos, Phase	0 um
FOV Pos, Slice	0 um

\\Research\Investigators\Frederick\allcustomer_20260909\ZPL_RG_EPSI_SE_v1b
TA: 3:28 min Coil Selection: Auto Voxel Size: 15.0×4.0×4.5 mm³ Rel. SNR: 1.00

Properties

Start measurement without further preparation	On
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
Graphic segment	Default
Inline Movie	Off

Routine

Position	Isocenter
Orientation	Transversal
Rotation	0.00 deg
FOV A >> P	240 mm
FOV R >> L	240 mm
FOV F >> H	72 mm
Vol A >> P	240 mm
Vol R >> L	240 mm
Vol F >> H	60 mm
TR	800.0 ms
TE	1.60 ms
Averages	1

Contrast - Common

TR	800.0 ms
TE	1.60 ms
Ernst angle	56 deg
WaterSupp Angle	90 deg
FOV Pos, Read	0 um
FOV Pos, Phase	0 um
FOV Pos, Slice	0 um
Flip Angle	90 deg
Preparation Scans	2
Averages	1
Water Suppression	
Water Suppr. BW	80 Hz

Resolution - Common

FOV A >> P	240 mm
FOV R >> L	240 mm
FOV F >> H	72 mm
Scan Res. A >> P	16
Scan Res. R >> L	60
Scan Res. F >> H	16
Interpol. Res. A >> P	128

Resolution - Common

Interpol. Res. R >> L	128
Interpol. Res. F >> H	64
Hamming	Off
Dimension	3D
Phase Encoding	Full
Normalize	Off

Geometry - Common

Position	Isocenter
Orientation	Transversal
Rotation	0.00 deg
FOV A >> P	240 mm
FOV R >> L	240 mm
FOV F >> H	72 mm
Vol A >> P	240 mm
Vol R >> L	240 mm
Vol F >> H	60 mm
Fully Excited Vol	Off

Geometry - AutoAlign

Slab Group	1
Position	Isocenter
Orientation	Transversal
Phase Encoding Dir.	A >> P
AutoAlign	---
Initial Position	Isocenter
L	0.0 mm
P	0.0 mm
H	0.0 mm
Initial Orientation	Transversal
Initial Rotation	0.00 deg

System - Miscellaneous

Coil Selection	Auto Coil Select
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Advanced
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off
Adj. Water Suppr.	Off

System - Adjust Volume

Position	Isocenter
Orientation	Transversal
Rotation	0.00 deg
A >> P	240 mm
R >> L	240 mm
F >> H	60 mm
Reset	Off

System - pTx

B1 Shim	TrueForm

System - Tx/Rx

Frequency 1H	123.249442 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Sequence - Common

Sequence Name	ZPL_epsi_se_v1b
Preparation Scans	2
Delta Frequency	-2.40 ppm
Dimension	3D
Save Uncombined	Off
Acquisition Delay	2.30 ms
Bandwidth	167000 Hz
Acquisition Duration	0 ms
Remove Oversampling	Off

Sequence - Nuclei

TX/RX Nucleus	1H
TX/RX Delta Frequency	0 Hz
TX Nucleus	None
TX Delta Frequency	0 Hz

Sequence - Special

PE Samp Pattern	No Customized Sampling Pattern
Mode. Bipolar	1 #
Mode. YreadNav	0 #
Mode: RFspoil	1 #
Ramp. Samp. EPSI	0 #
Ramping time (RO)	110 us
Ramping time (PE)	120 us
Grad duration (PE)	600 us
Blip ramp time (PE)	60 us
Blip ramp time (SL)	60 us
Grad. EchoSpace(RO)	1160 us
Grad. flat time(RO)	360 us
Readout ADC (RO)	360 us
ADC dwell time	6 us
Acq. advance time	0 us
Refocussing pulse voi	160 %
Crusher duration (Cr)	1200 us
Num: Echo-Shift	0 #

Sequence - Special

Number of TEs	2 #
TE values[1]	50 ms
TE values[2]	190 ms
TE values[3]	420 ms
TE values[4]	620 ms
TE values[5]	820 ms
TE values[6]	1020 ms
TE values[7]	1220 ms
TE values[8]	1420 ms
Echo-Pairs. EPSI[1]	54 #
Echo-Pairs. EPSI[2]	256 #
Echo-Pairs. EPSI[3]	40 #
Echo-Pairs. EPSI[4]	40 #
Echo-Pairs. EPSI[5]	40 #
Echo-Pairs. EPSI[6]	40 #
Echo-Pairs. EPSI[7]	40 #
Echo-Pairs. EPSI[8]	40 #
Echos of FID	14 #
Echos before TEs[1]	16 #
Echos before TEs[2]	54 #
Echos before TEs[3]	0 #
Echos before TEs[4]	0 #
Echos before TEs[5]	0 #
Echos before TEs[6]	0 #
Echos before TEs[7]	0 #
Echos before TEs[8]	0 #
Crusher amplitude1(Cr)[1]	16 us
Crusher amplitude1(Cr)[2]	8 us
Crusher amplitude1(Cr)[3]	16 us
Crusher amplitude1(Cr)[4]	16 us
Crusher amplitude1(Cr)[5]	16 us
Crusher amplitude1(Cr)[6]	16 us
Crusher amplitude1(Cr)[7]	16 us
Crusher amplitude1(Cr)[8]	16 us
Fnav. read points	100 #
Fnav. dwell time	10 us